MIL-M-38510/2E
24 December 1974
SUPERSEDING
MIL-M-0038510/2D(USAF)
15 October 1973 and
MIL-M-38510/2C
7 November 1972

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, TTL, FLIP-FLOPS, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, TTL, bistable logic microcircuits. Three product assurance classes and a choice of case outline/lead finish are provided for each type and are reflected in the complete part number.

1.2 Part number. The complete part number shall be as shown in the following example:

M38510	/002	01	B	A	C
Military designator	Detail specification	Device type (1.2.1)	Device class (1.2.2)	Case outline (1.2.3)	Lead finish (3.3)

1.2.1 Device type. Device type shall be as shown in the following:

Device type	Circuit
01	Single J-K master-slave flip-flop
02	Dual J-K master-slave flip-flop, no preset
03	Dual J-K master-slave flip-flop, no preset
04	Dual J-K master-slave flip-flop
05	Dual D-type edge-triggered flip-flop
06	Single edge-triggered J-K flip-flop
07	Dual D-type edge-triggered flip-flop, buffered output

1.2.2 Device class. Device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Letter	Case outline, MIL-M-38510, appendix C
A	F-1 (14-pin, 1/4" x 1/4", flat pack)
B	F-3 (14-pin, 1/8" x 1/4", flat pack)
C	D-1 (14-pin, 1/4" x 3/4", dual-in-line)
D	F-2 (14-pin, 1/4" x 3/8", flat pack)
E	D-2 (16-pin, 1/4" x 7/8", dual-in-line)
F	F-5 (16-pin, 1/4" x 3/8", flat pack)

1.2.4 Absolute maximum ratings.

Supply voltage range	-0.5 to 7.0 Vdc
Input voltage range	-1.5 Vdc at -12 mA to 5.5 Vdc
Storage temperature range	-65° C to 150° C
Maximum power dissipation per flip-flop, P_D	110 mW 1/
Lead temperature (soldering, 10 seconds)	300° C
Thermal resistance, junction to case	θ_{JC} = 0.09° C/mW for flat packs; 0.08° C/mW for dual-in-line pack
Junction temperature	T_J = 175° C

1/ Must withstand the added P_D due to short circuit condition (e.g. I_{OS}) at one output for 5 seconds duration.

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1.2.5 Recommended operating conditions.

Supply voltage	4.5 Vdc minimum to 5.5 Vdc maximum
Minimum high level input voltage	2.0 Vdc
Maximum low level input voltage	0.8 Vdc
Normalized fanout (each output) 1/	10 maximum
Ambient operating temperature range	-55° C to 125° C
Input set up time	
Device type	
01, 02, 03, 04	$\geq$ clock pulse width
05, 06, 07	20 ns minimum
Input hold time	
Device type	
01, 02, 03, 04,	0
05, 06, 07	5 ns minimum

2. APPLICABLE DOCUMENT

2.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein. In the event of conflict between MIL-M-38510 and this detail specification, this detail specification shall govern.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510, and herein.

3.2.1 Logic diagram and terminal connections. The logic diagram and terminal connections shall be as specified on figure 1.

3.2.2 Truth tables and logic equations. The truth tables and logic equations shall be as specified on figure 2.

3.2.3 Schematic circuit. The schematic circuit shall be as specified on figure 3.

3.2.4 Case outlines. Case outlines shall be as specified in MIL-M-38510 and in 1.2.3 herein.

3.3 Lead material and finish. Lead material and finish shall be in accordance with MIL-M-38510.

3.4 Electrical performance characteristics. The electrical performance characteristics are specified in table I, and apply over the full recommended ambient operating temperature range, unless otherwise specified.

3.5 Rebonding. Rebonding shall be in accordance with MIL-M-38510.

1/ Device will fanout in both high and low levels to the specified number of I_{IL1}/I_{IH1} inputs of the same device type as that being tested.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions 9/	Device type	Limits		Units
				Min	Max	
High-level output voltage	V_{OH}	V_{CC} = 4.5 V, I_{OH} = -400 µA	All	2.4	---	Volts
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V, I_{OL} = 16 mA	All		0.4	Volts
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{OL} = -12 mA, T_A = 25°C	All		-1.5	Volts
Low-level input current	I_{IL1}	V_{CC} = 5.5 V, V_{IN} = 0.4 V 1/	01, 02, 03, 04, 05, 06	-0.7	-1.6	mA
			07	-0.5	-1.6	mA
Low-level input current	I_{IL2}	V_{CC} = 5.5 V, V_{IN} = 0.4 V 2/	01, 02, 03, 04, 05	-1.4	-3.2	mA
			07	-1.0	-3.2	mA
Low-level input current	I_{IL3}	V_{CC} = 5.5 V, V_{IN} = 0.4 V 6/	01, 02, 03, 04	-0.7	-3.2	mA
High-level input current	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.4 V 5/	All	0	40	µA
High-level input current	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 5.5 V 5/	All	0	100	µA
High-level input current	I_{IH3}	V_{CC} = 5.5 V, V_{IN} = 2.4 V 3/	All	0	80	µA
High-level input current	I_{IH4}	V_{CC} = 5.5 V, V_{IN} = 5.5 V 3/ 7/	All	0	200	µA
High-level input current	I_{IH5}	V_{CC} = 5.5 V, V_{IN} = 2.4 V 7/ 8/	01, 02, 03, 04, 05, 07	-850 0	-50 120	µA µA
High-level input current	I_{IH6}	V_{CC} = 5.5 V, V_{IN} = 5.5 V 8/	05, 07	0	300	µA
Short-circuit output current	I_{OS}	V_{CC} = 5.5 V, V_{IN} = 0 4/	All	-20	-57	mA
Supply current per device	I_{CC}	V_{CC} = 5.5 V, V_{IN} = 5 V	01		20	mA
			02, 03, 04		40	
			05, 06, 07		30	
Maximum clock frequency	f_{MAX}	V_{CC} = 5 V, CL = 50 pF minimum, RL = 390 Ω ±5%	01, 02, 03, 04, 05, 07	5		MHz
			06	7.5		
Propagation delay to high logic level (clear or preset to output)	t_{PLH}		01, 02, 03, 04, 05	5	39	ns
			06	5	62	
			07	5	31	
Propagation delay to low logic level (clear or preset to output)	t_{PHL}		01, 02, 03, 04, 05	5	50	ns
			06	5	62	
			07	5	39	
Propagation delay to high logic level (clock to output)	t_{PLH}		06	5	62	ns
			01, 02, 03, 04, 05	5	39	
			07	5	31	
Propagation delay to low logic level (clock to output)	t_{PHL}		06	5	62	ns
			01, 02, 03, 04, 05	5	50	
			07	5	39	

1/ Input condition - J or K for device types 01, 02, 03, 04, 06, and preset or D for device types 05 and 07, and clock, clear or preset for device type 06.
2/ Input condition - Clock for device types 01, 02, 03 and 04, and clear or clock for device types 05 and 07.
3/ Input condition - Clear or preset for device types 01, 02, 03, 04, 05, 06 and 07 and clock for device types 05 and 07.
4/ Nor more than one output should be shorted at a time.
5/ Input condition - J or K for device types 01, 02, 03, 04, 06, and D for device types 05 and 07, and clock for device type 06.
6/ Input condition - Clear or preset for device types 01, 02, 03 and 04.
7/ Input condition - Clock for device types 01, 02, 03, and 04.
8/ Input condition - Clear for device types 05 and 07.
9/ See table III for complete terminal conditions.

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3.6 Electrical test requirements. Electrical test requirements shall be as specified in table III for the applicable device type and device class. The subgroups of table III which constitute the minimum electrical test requirements for screening, qualification, and quality conformance by device class are specified in table II. (Subgroups 7 and 8 testing requires only a summary of attributes data.)

TABLE II. Electrical test requirements.

MIL-STD-883 Test requirements	Subgroups (see table III)		
	Class A devices	Class B devices	Class C devices
Interim electrical parameters (Pre Burn-In) (Method 5004)	1	1	Not reqd
Final electrical test parameters (Method 5004)	1*, 2, 3, 7, 8, 9	1*, 2, 3, 7, 9	1, 7
Group A test requirements (Method 5005)	1, 2, 3, 7, 8, 9, 10, 11	1, 2, 3, 7, 9	1, 2, 3, 7, 9
Group C end point electrical parameters (Method 5005)	1, 2, 3	1, 2, 3	1
Additional electrical subgroups for Group C periodic inspections	None	10, 11	None

* PDA applies to subgroup 1 (see 4.3(h)).

3.7 Marking. Marking shall be in accordance with MIL-M-38510 and 1.2 herein. At the option of the manufacturer, the following marking may be omitted from the body of the microcircuit, but shall be retained on the initial container:

(a) Country of origin.

4. PRODUCT ASSURANCE PROVISIONS.

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and Method 5005 of MIL-STD-883 except as modified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified herein for groups A, B, and C inspections (see 4.4.1, 4.4.2, and 4.4.3). After qualification of one or more electrically and structurally similar types with a single lead finish, other lead finishes of the same case outline may be qualified by submitting a single type in the qualified case outline to the group B, subgroup 3 test and the group C, subgroup 1, 3, and 4 test.

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4.3 Screening. Screening shall be in accordance with Method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

(a) Test samples for the group B bond strength test specified in Method 5005 of MIL-STD-883 may, at the manufacturer's option, be randomly selected immediately following the internal visual (precap) inspection and prior to sealing (see 4.4.2(b)).
(b) Temperature cycling (Method 1010 of MIL-STD-883).
    (1) Omit seal test as post-test measurement.
(c) Thermal shock (Method 1011 of MIL-STD-883), when substituted for temperature cycling.
    (1) Omit seal test as post-test measurement.
(d) Burn-in test (Method 1015 of MIL-STD-883).
    (1) Test condition D or E, using the circuit shown on figure 4, or equivalent.
    (2) T_A = 125° C minimum.
(e) Reverse bias burn-in and interim electrical test in accordance with 3.1.10 of Method 5004 of MIL-STD-883 may be omitted.
(f) Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.
(g) External visual inspection shall not include measurement of case and lead dimensions.
(h) Percent defective allowable (PDA) - The PDA is specified as 5 percent for class A devices and 10 percent for class B devices based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of Group A, subgroup 1 after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510.

4.4.1 Group A inspection. Group A inspection shall consist of the test subgroups and LTPD values shown in table I of Method 5005 of MIL-STD-883 and as follows:

(a) Tests shall be as specified in table II.
(b) Subgroups 4, 5, 6, 7, and 8 of table I of Method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall consist of the test subgroups and LTPD values shown in table II of Method 5005 of MIL-STD-883 and as follows:

(a) Bond strength test may be conducted on samples collected prior to sealing (see 4.3(a)).

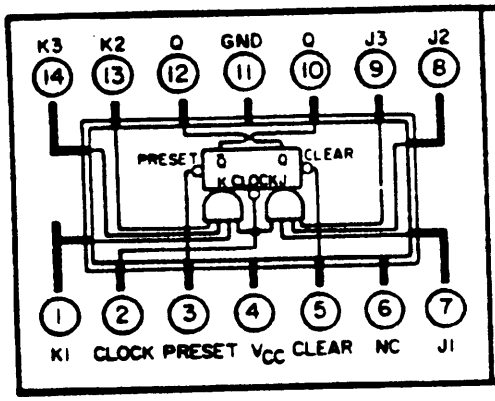
4.4.3 Group C inspection. Group C inspection shall consist of the test subgroups and LTPD values shown in table III of Method 5005 of MIL-STD-883 and as follows:

(a) End point electrical parameters shall be as specified in table II.
(b) Subgroups 7 and 8 shall be added to the group C inspection requirements for class B devices and shall consist of the tests, conditions, and limits specified for subgroups 10 and 11 of group A.

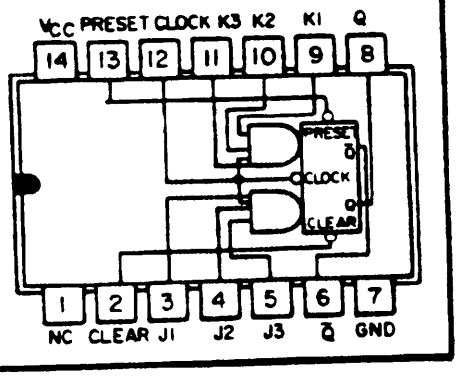
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Device type 01

Cases A, B, and D

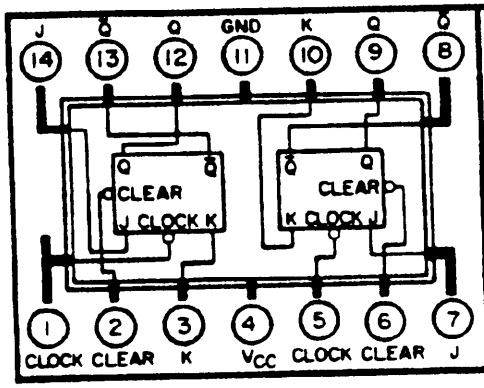


Case C

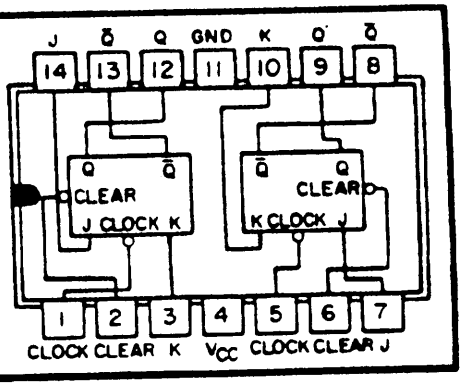


Device type 02

Cases A, B, and D



Case C



Device type 03

Case C

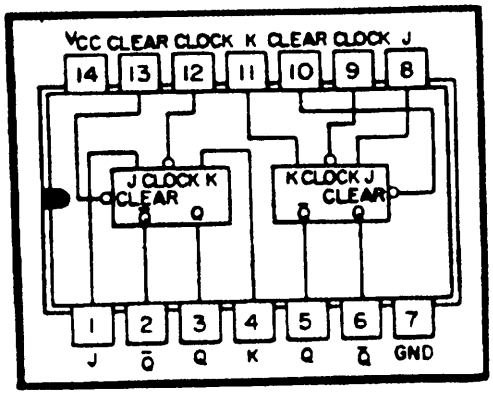
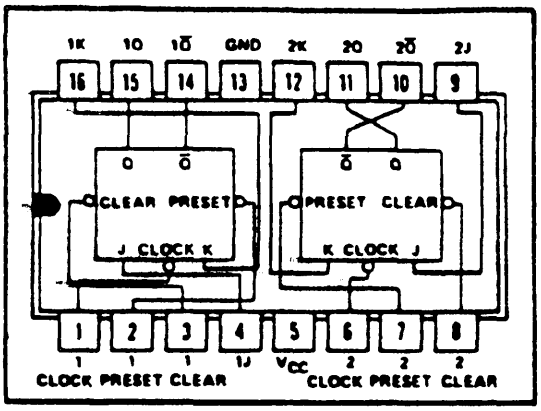


FIGURE 1. Logic diagram and terminal connections.

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Device type 04

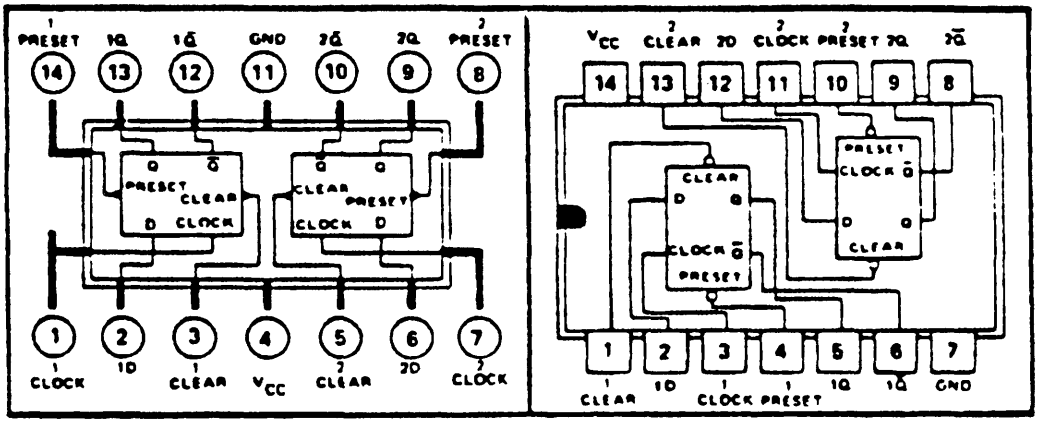
Cases E and F



Device types 05 and 07

Cases A, B, and D

Case C



Device type 06

Cases A, B, and D

Case C

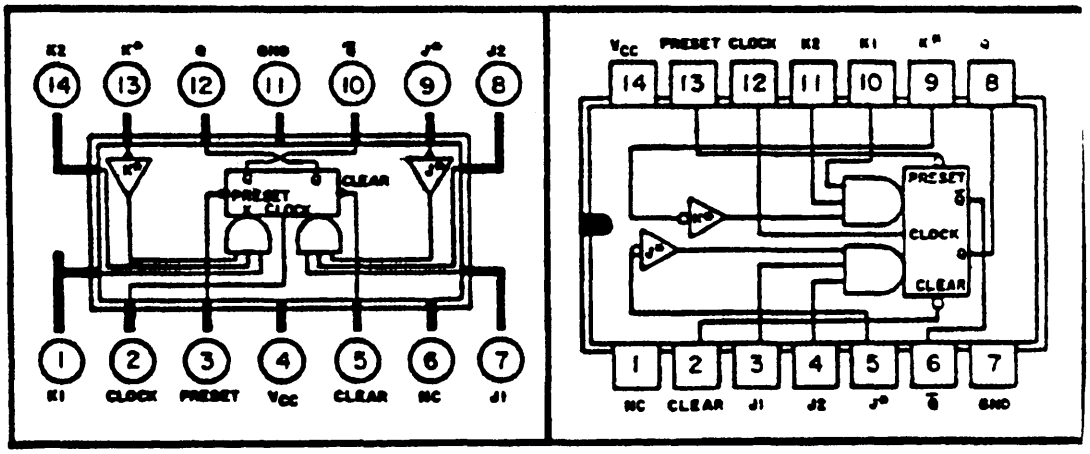


FIGURE 1. Logic diagram and terminal connections - Continued.

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Device type 01

Truth table		
t_n		t_n+1
J	K	Q
L	L	Q_n
L	H	L
H	L	H
H	H	$\overline{Q}_n$

Positive logic: Low input to preset sets Q to high-level
Low input to clear sets Q to low-level
Preset and clear are independent of clock and dominate regardless of the state of clock or J or K inputs.

NOTES: 1. J = J1 · J2 · J3
2. K = K1 · K2 · K3
3. t_n = Bit time before clock pulse.
4. t_n+1 = Bit time after clock pulse.

Device types 02 and 03

Truth table each flip-flop		
t_n		t_n+1
J	K	Q
L	L	Q_n
L	H	L
H	L	H
H	H	$\overline{Q}_n$

Positive logic: Low input to clear sets Q to low-level
Clear is independent of clock and dominate regardless of the state of clock or J or K inputs.

NOTES: 1. t_n = Bit time before clock pulse.
2. t_n+1 = Bit time after clock pulse.

FIGURE 2. Truth tables.

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Device type 04

Truth table each flip-flop		
t_n		t_n+1
J	K	Q
L	L	Q_n
L	H	L
H	L	H
H	H	$\overline{Q}_n$

Positive logic: Low input to preset sets Q to high-level.
Low input to clear sets Q to low-level.
Preset and clear are independent of clock and dominate regardless of the state of clock or J or K inputs.

NOTES: 1. t_n = Bit time before clock pulse.
2. t_n+1 = Bit time after clock pulse.

Device types 05 and 07

Truth table each flip-flop		
t_n	t_n+1	
INPUT D	OUTPUT Q	OUTPUT $\overline{Q}$
L	L	H
H	H	L

Positive logic: Low input to preset sets Q to high-level.
Low input to clear sets Q to low-level.
Clear and preset are independent of clock and dominate regardless of the state of clock or D input.

NOTES: 1. t_n = Bit time before clock pulse.
2. t_n+1 = Bit time after clock pulse.

FIGURE 2. Truth tables - Continued.

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Device type 06

Truth table		
t_n		t_{n+1}
J	K	Q
L	L	Q_n
L	H	L
H	L	H
H	H	$\overline{Q}_n$

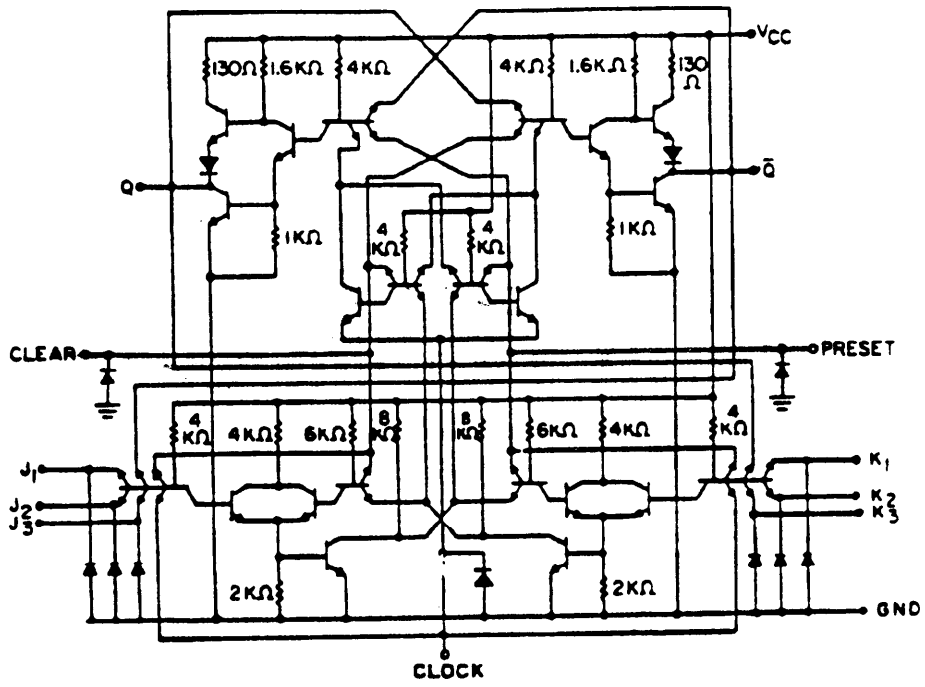
Positive logic: Low input to preset sets Q to high-level.
Low input to clear sets Q to low-level.
Preset or clear function can occur only when clock input is low.

NOTES:
1. $J = J1 \cdot J2 \cdot \overline{J*}$
2. $K = K1 \cdot K2 \cdot \overline{K*}$
3. t_n = Bit time before clock pulse.
4. t_{n+1} = Bit time after clock pulse.
5. If inputs J* or K* are not used they must be grounded.

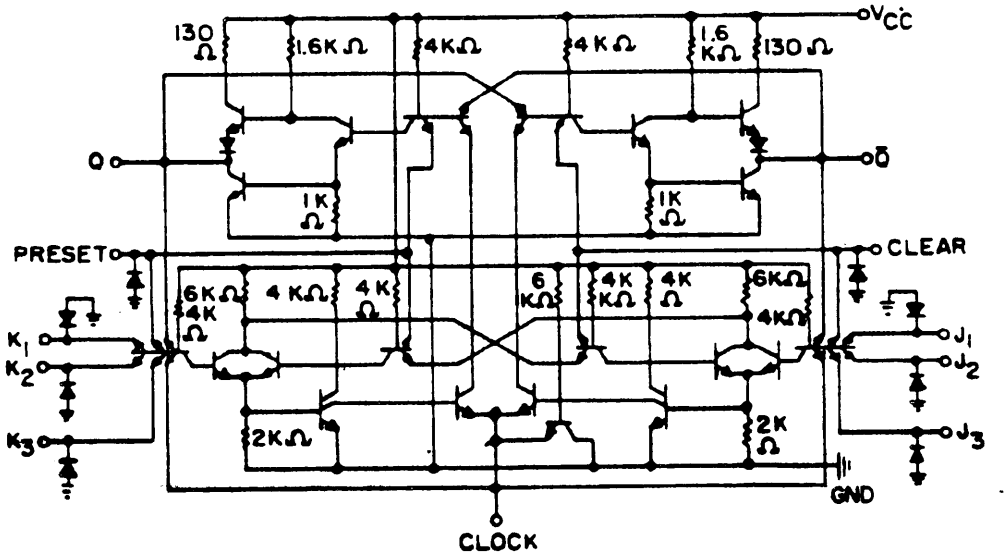
FIGURE 2. Truth tables - Continued.

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Device type 01



CIRCUIT A

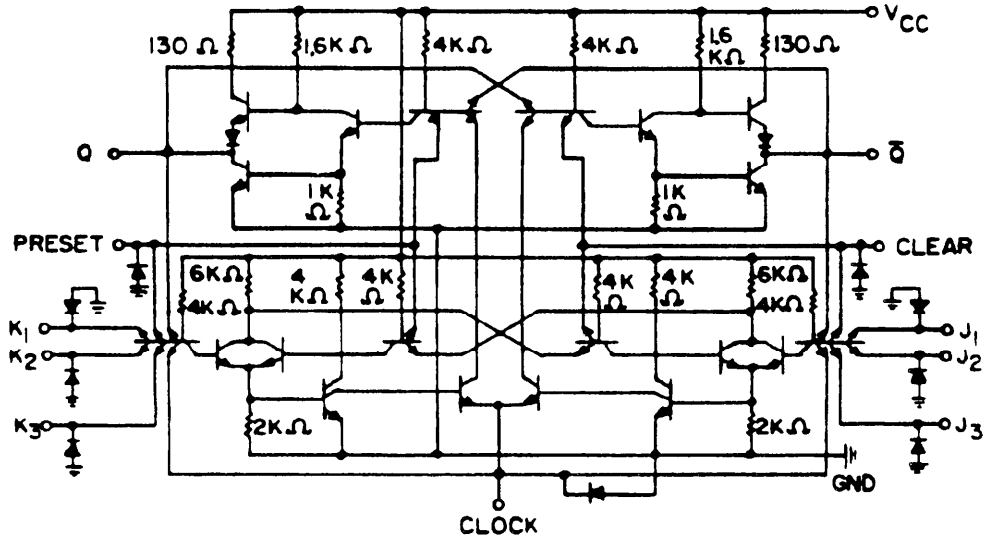


CIRCUIT B

FIGURE 3. Schematic circuits.

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Device type 01 - Continued.



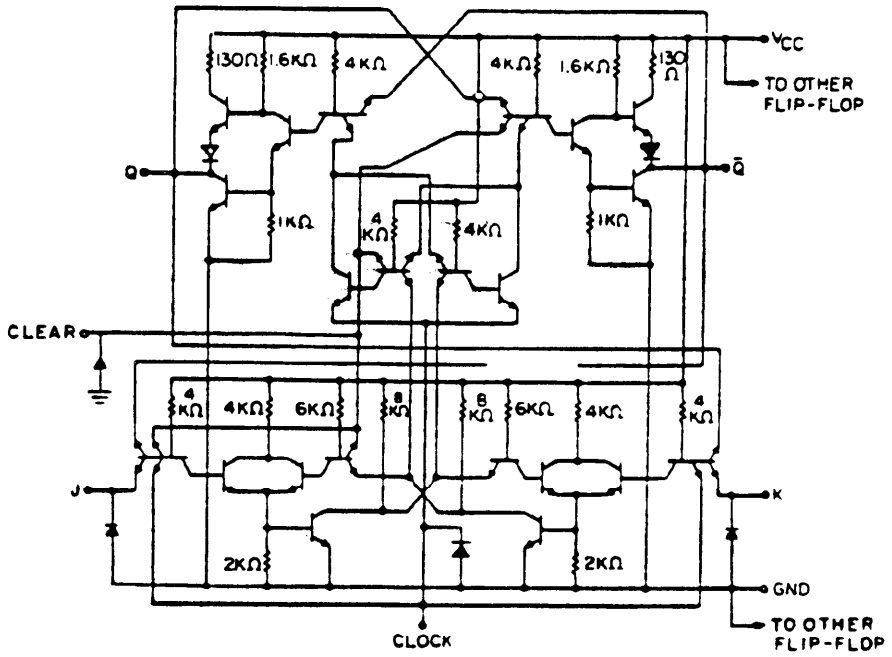
CIRCUIT C

NOTES:
1. Circuits A, B and C are the only acceptable variations for device type 01.
2. All resistance values shown are nominal.

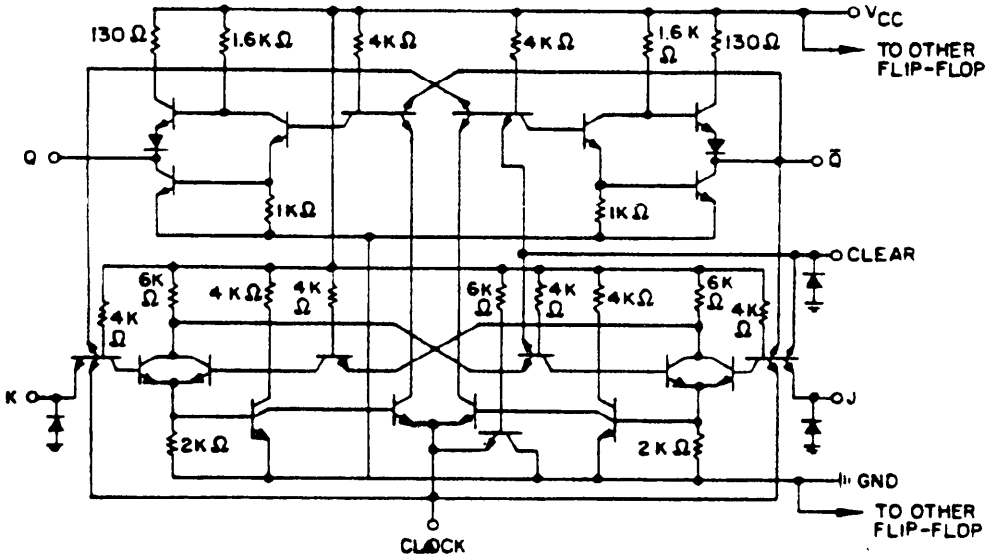
FIGURE 3. Schematic circuits - Continued.

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Device types 02 and 03



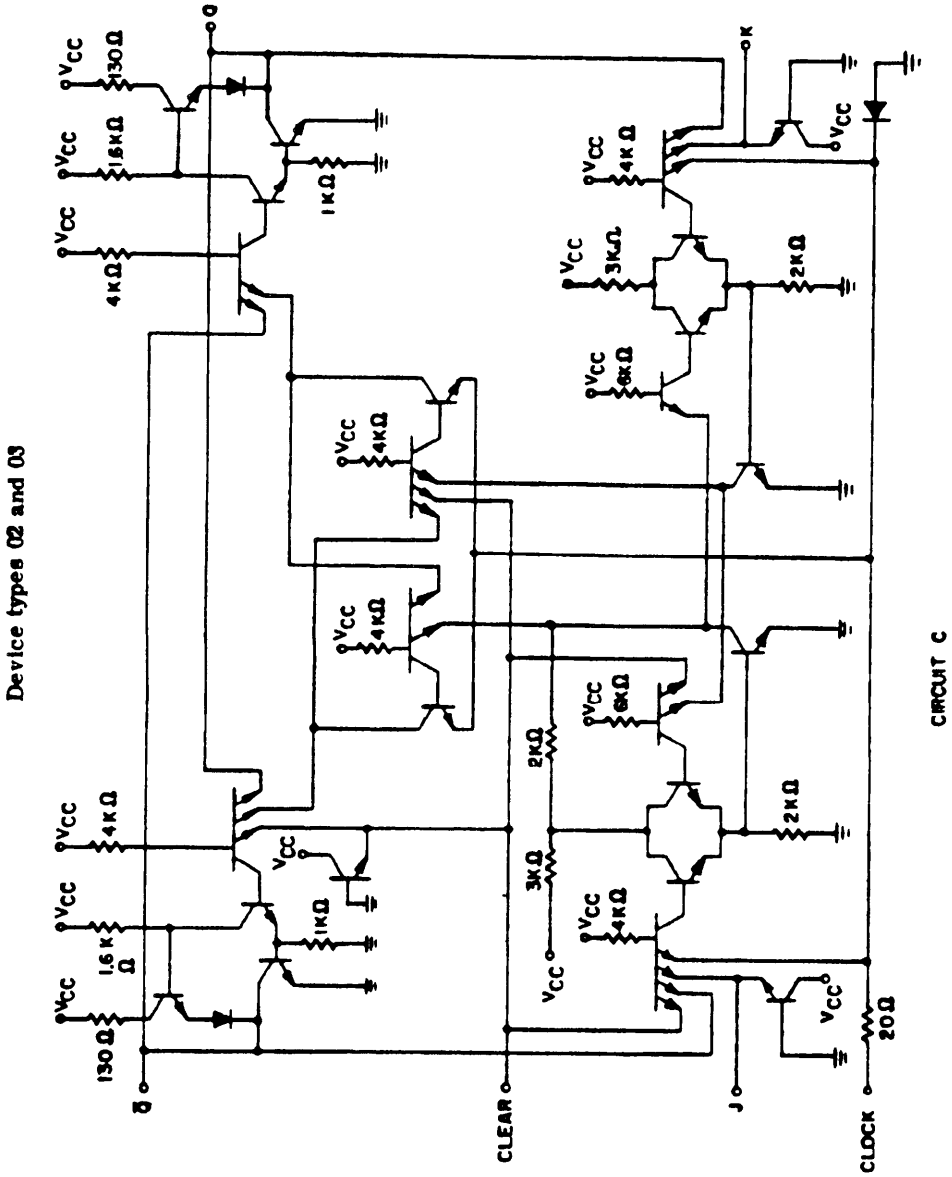
CIRCUIT A



CIRCUIT B

FIGURE 3. Schematic circuits - Continued.

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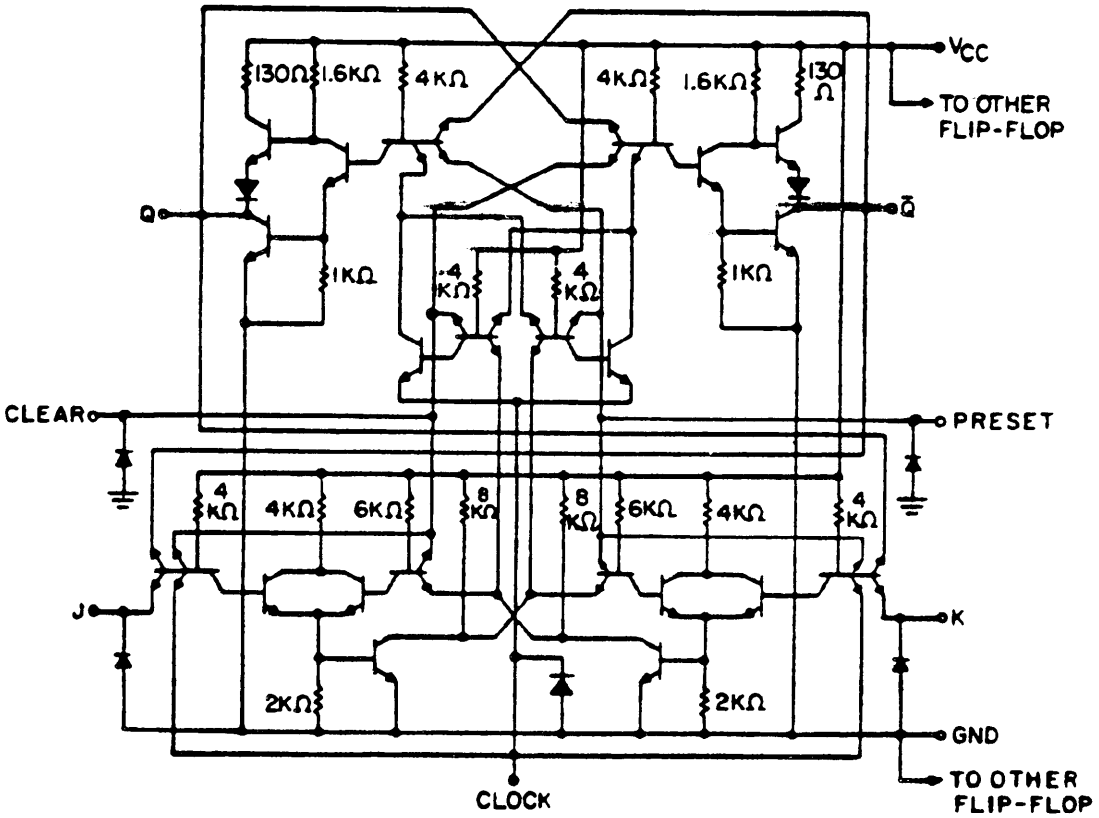


NOTES:
1. Circuits A, B and C are the only acceptable variations for device types 02 and 03.
2. All resistance values shown are nominal.

FIGURE 3. Schematic circuits - Continued.

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Device type 04



CIRCUIT A

FIGURE 3. Schematic circuits - Continued.

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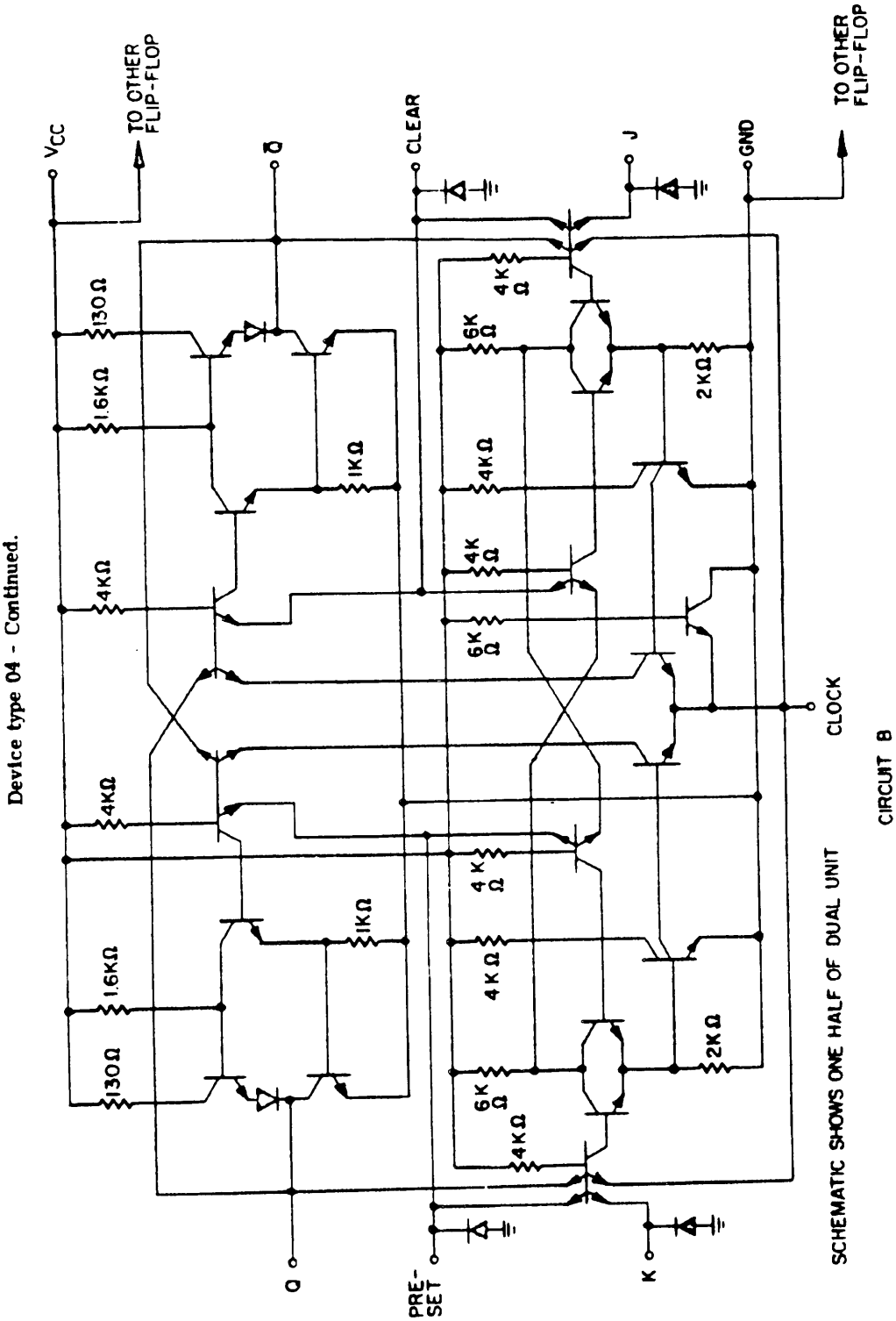
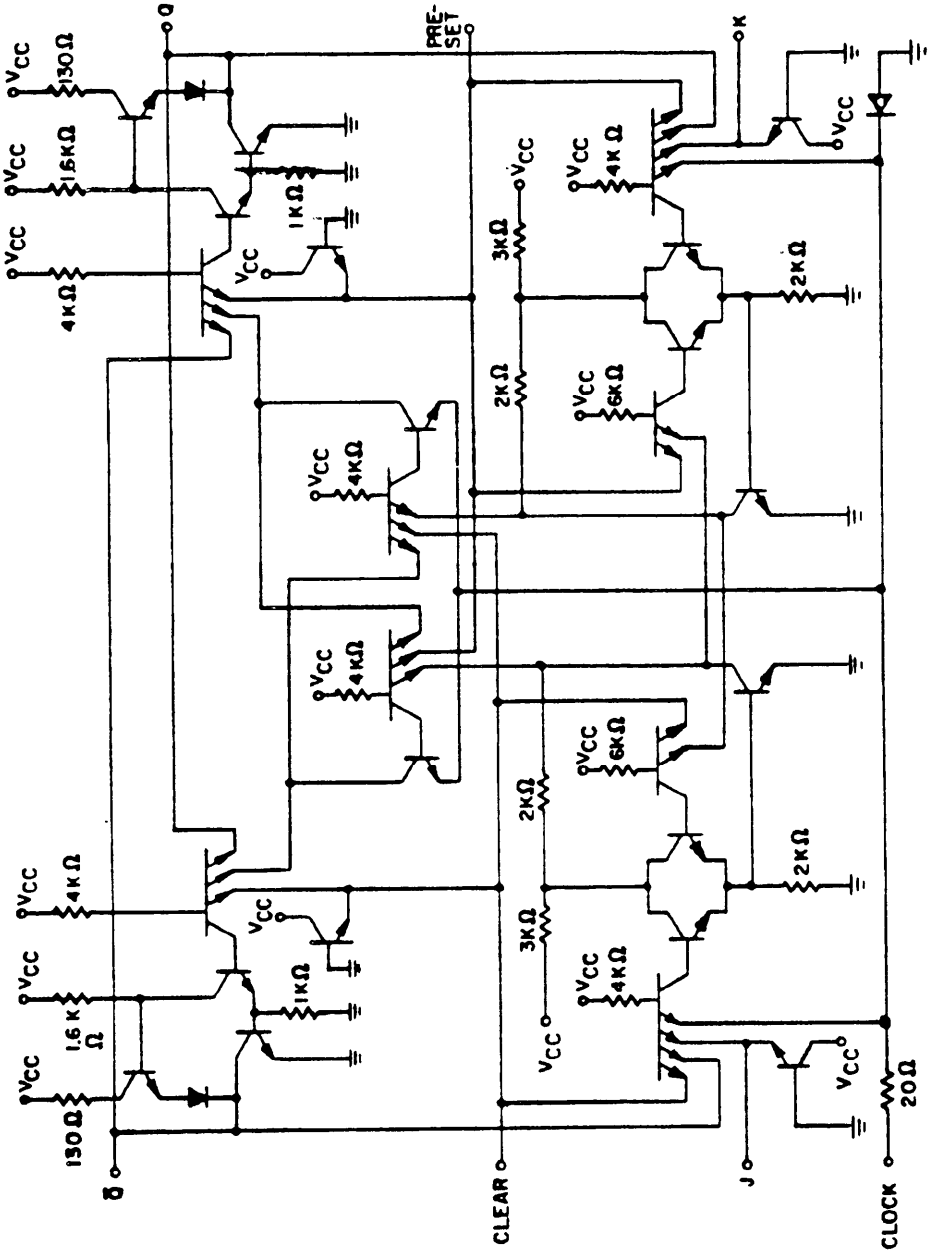


FIGURE 3. Schematic circuits - Continued.

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Device type 04 - Continued.



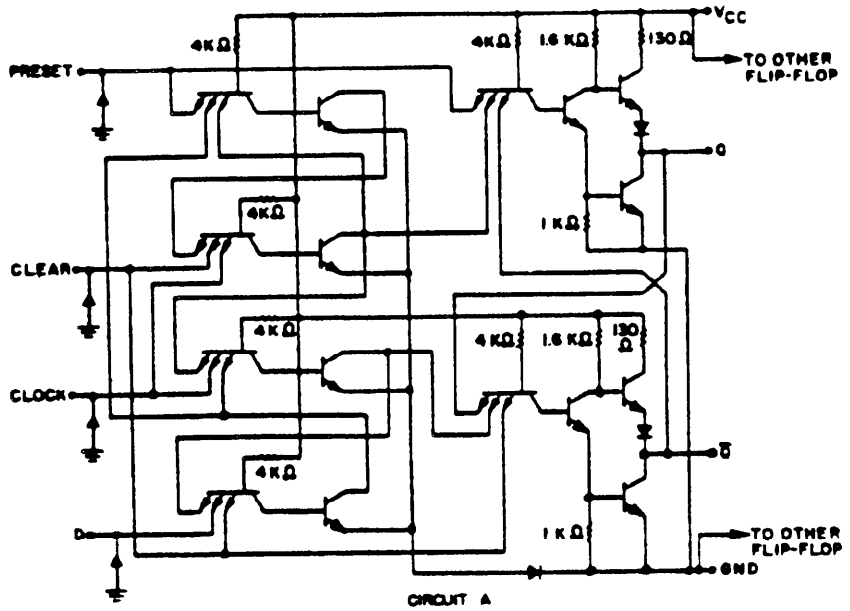
CIRCUIT C

NOTES:
1. Circuits A, B and C are the only acceptable variations for device type 04.
2. All resistance values shown are nominal.

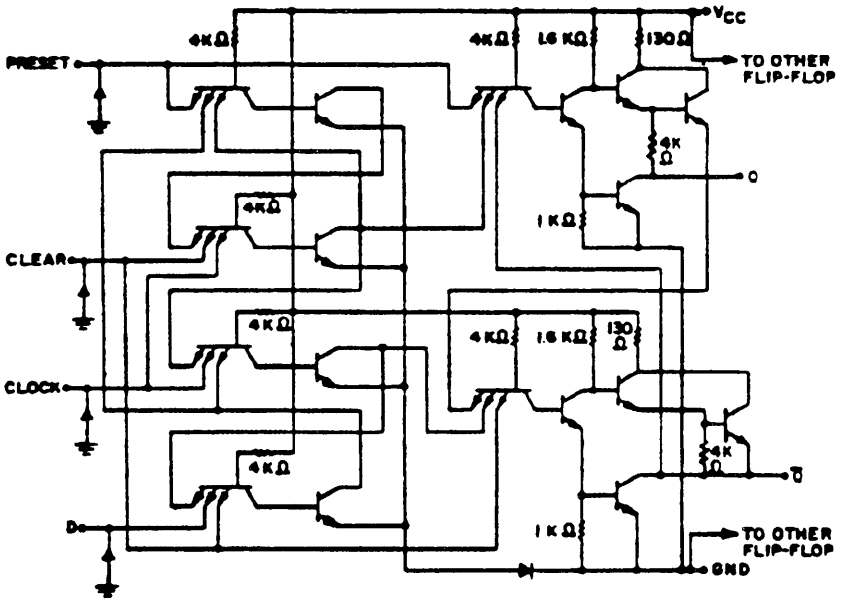
FIGURE 3. Schematic circuits - Continued.

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Device type 05



CIRCUIT A



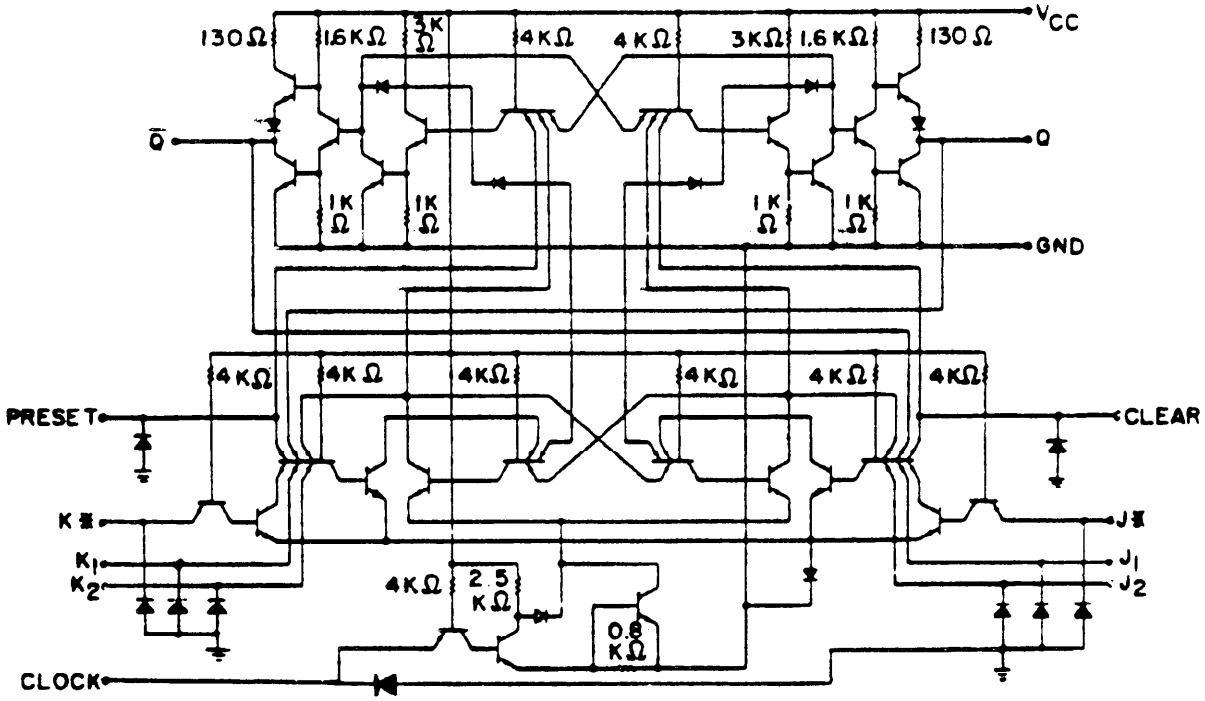
CIRCUIT B

NOTES:
1. Circuits A and B are the only acceptable variations for device type 05.
2. All resistance values shown are nominal.

FIGURE 3. Schematic circuits - Continued.

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Device type 06

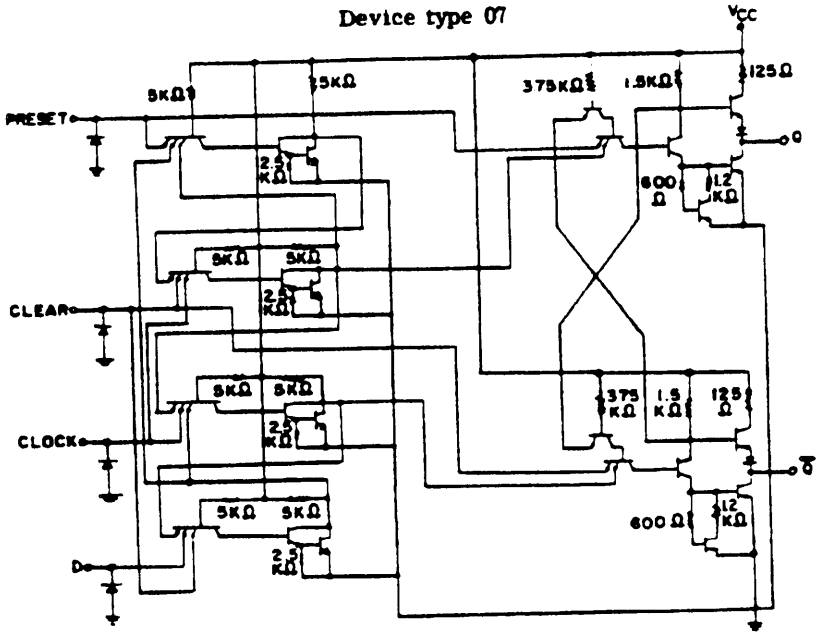


NOTE: All resistance values shown are nominal.

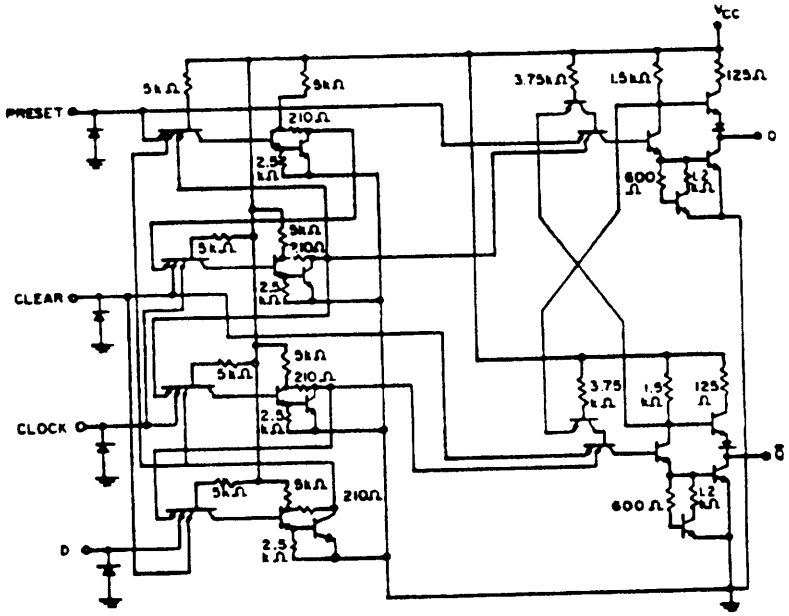
FIGURE 3. Schematic circuits - Continued.

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Device type 07



CIRCUIT A



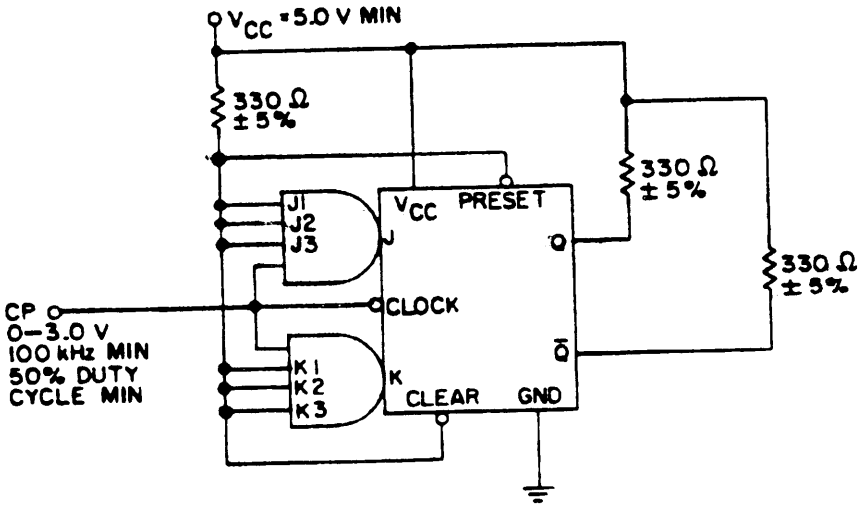
CIRCUIT B

NOTES:
1. Circuits A and B are the only acceptable variations for device type 07.
2. All resistance values shown are nominal.

FIGURE 3. Schematic circuits - Continued.

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Device type 01



Device types 02 and 03

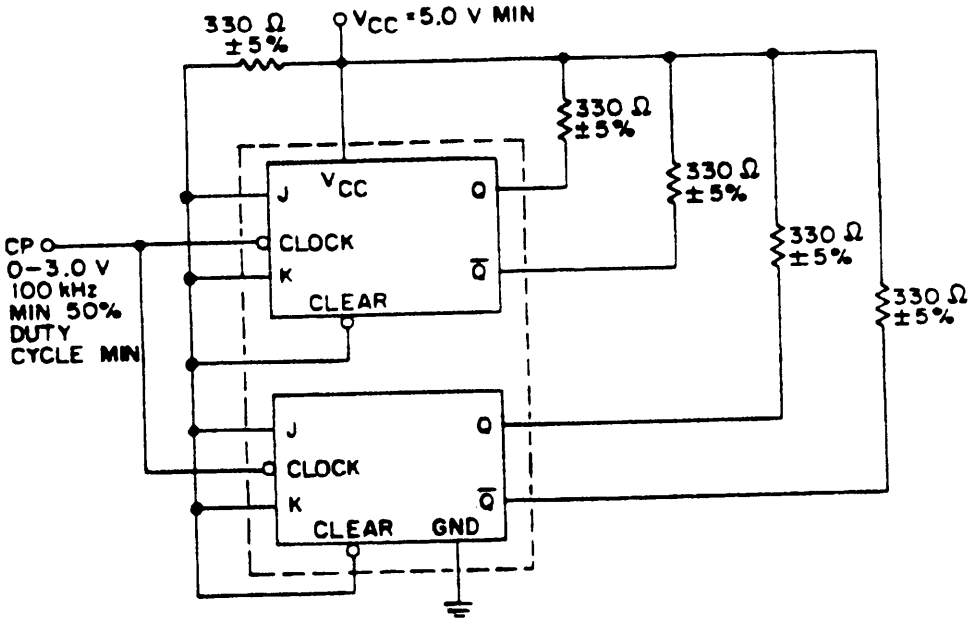
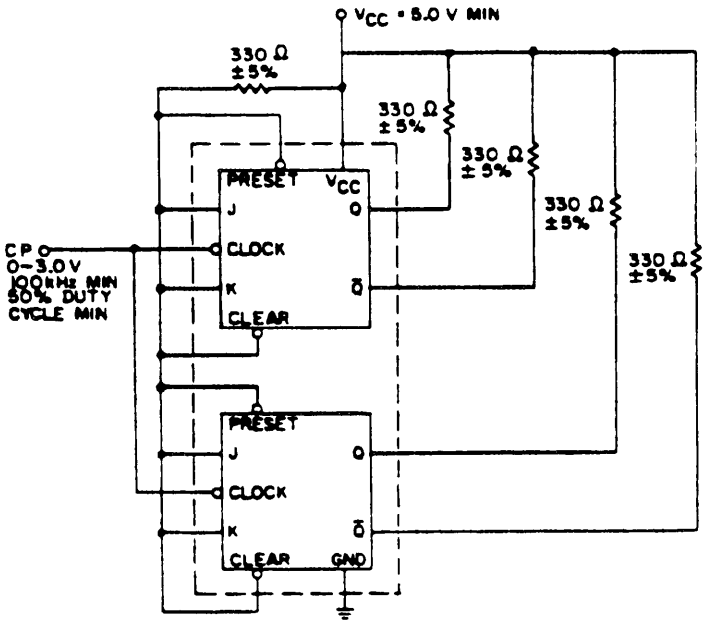


FIGURE 4. Burn-in and life-test circuits.

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Device type 04



Device types 05 and 07

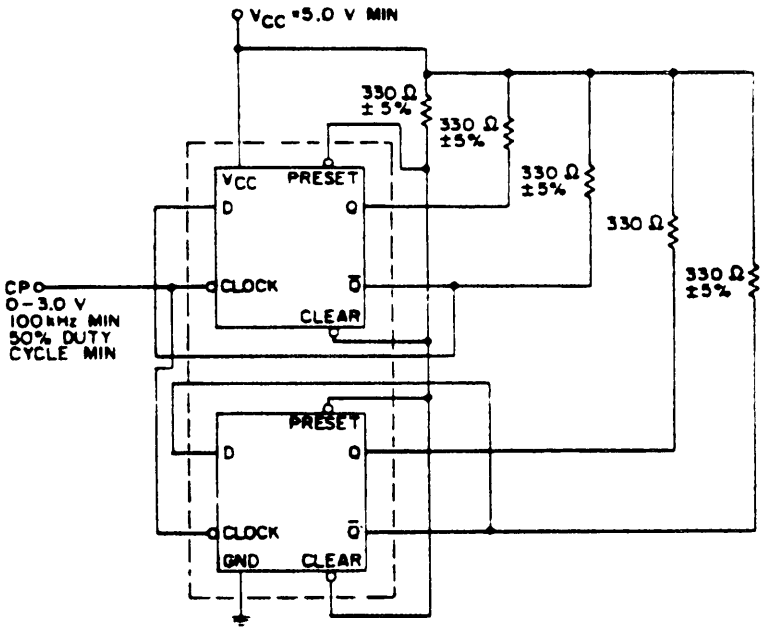


FIGURE 4. Burn-in and life-test circuits - Continued.

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Device type 06

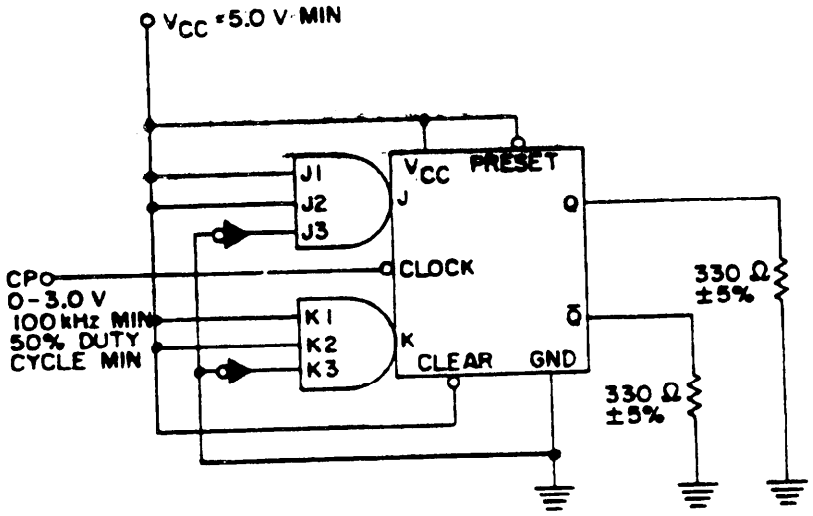
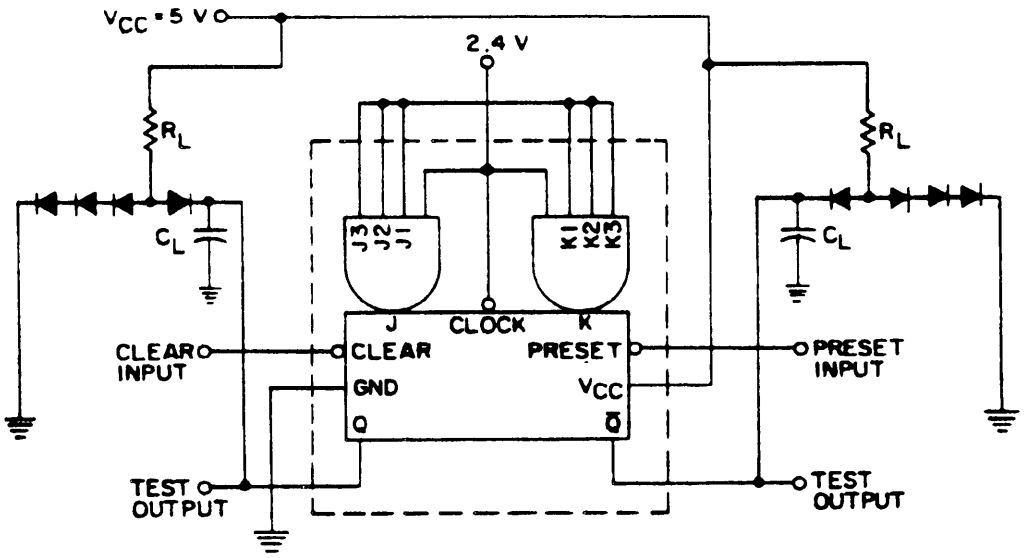
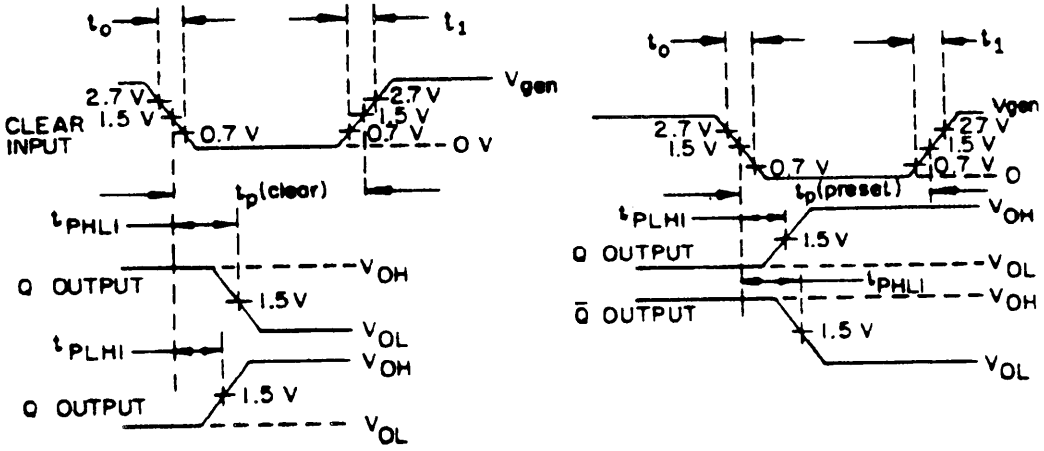


FIGURE 4. Burn-in and life-test circuits - Continued.

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TEST CIRCUIT

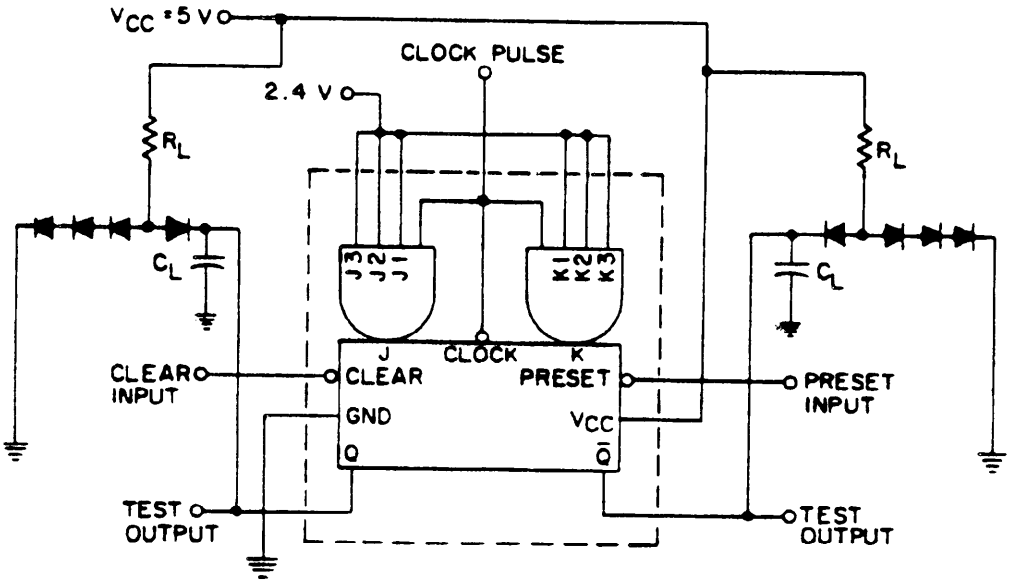


VOLTAGE WAVEFORMS

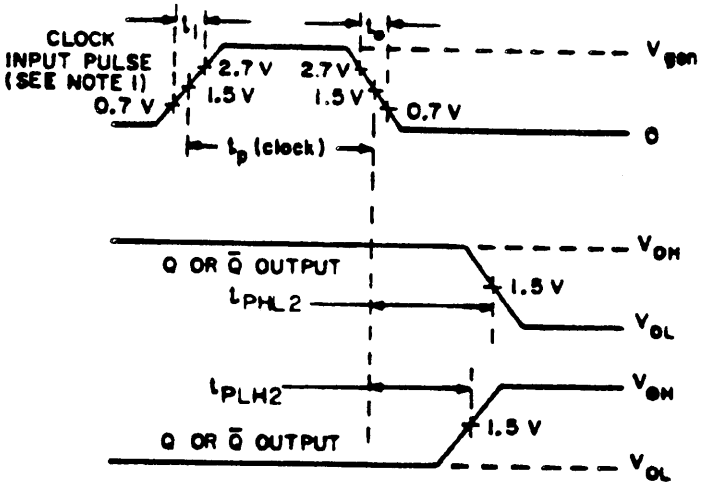
NOTES:
1. Clear or preset inputs dominate regardless of the state of clock or J-K inputs.
2. Clear or preset input pulse characteristics: Vgen = 3 V, $t_0 = t_1 = 10$ ns, t_p(clear) = t_p(preset) = 30 ns, PRR = 1 MHz, and $Z_{out} \approx 50\Omega$.
3. C_L = 50 pF, minimum (C_L includes probe and jig capacitance).
4. $R_L = 390\Omega \pm 5\%$.
5. All diodes are 1N3064, or equivalent.
6. When testing clear to output switching, preset input shall have a negative pulse. When testing preset output switching, clear shall have a negative pulse (see table III).

FIGURE 5. Clear and preset switching test circuit and waveforms for device type 01.

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TEST CIRCUIT

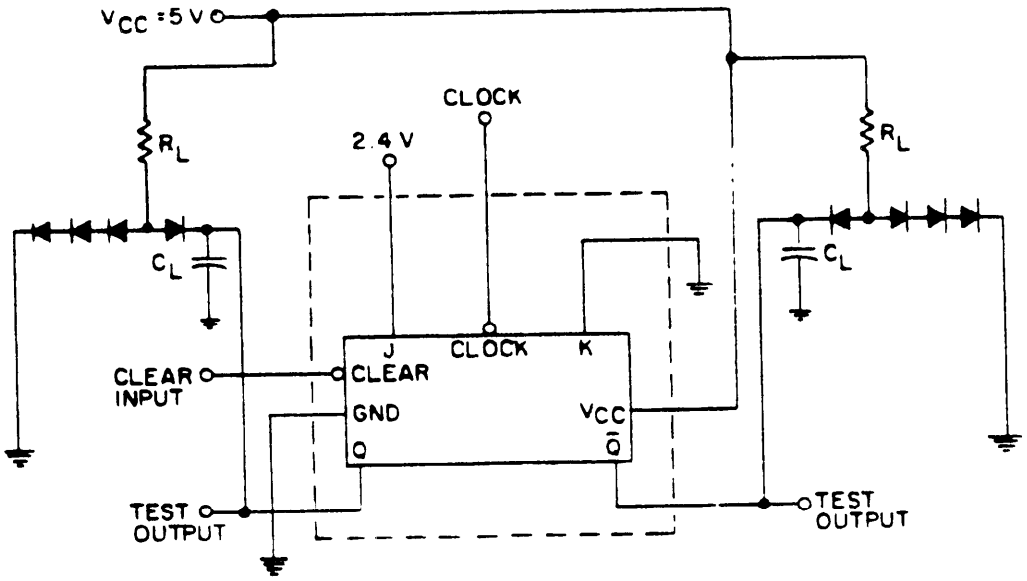


VOLTAGE WAVEFORMS

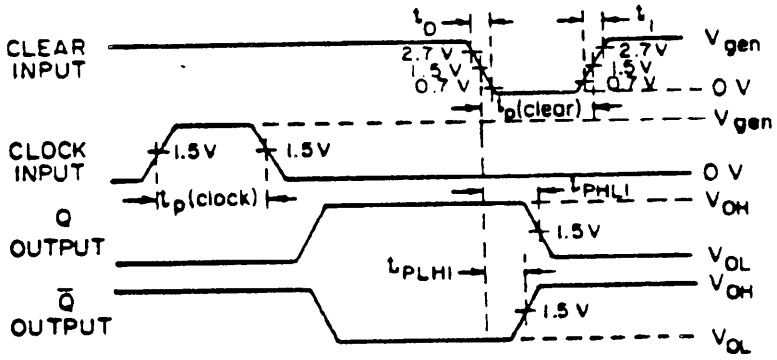
NOTES:
1. Clock input characteristics for t_{PLH}, t_{PHL} (clock to output), Vgen = 3 V, $t_1 = t_0 \leq 10$ ns, t_p (clock) = 25 ns, and PRR = 1 MHz. All J and K inputs are at 2.4 V. When testing f_{MAX} the clock input characteristics are Vgen = 3 V, $t_1 = t_0 \leq 10$ ns, t_p (clock) = 20 ns, and PRR = see table III.
2. J = J1 · J2 · J3; and K = K1 · K2 · K3.
3. All diodes are 1N3064, or equivalent.
4. C_L = 50 pF minimum (C_L includes probe and jig capacitance).
4. R_L = 390Ω ±5%.

FIGURE 6. Synchronous switching test circuit for device type 01.

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TEST CIRCUIT

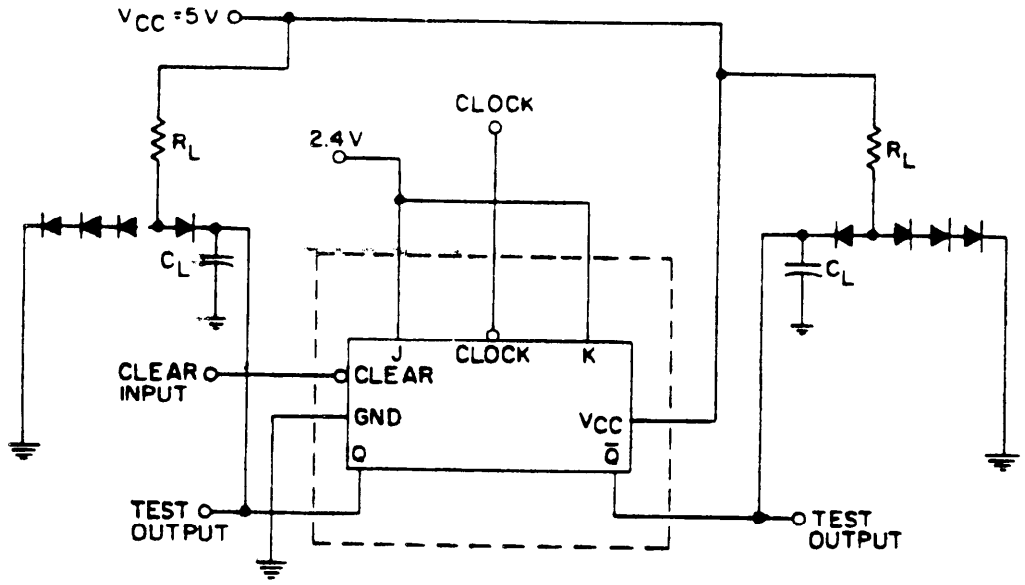


VOLTAGE WAVEFORMS

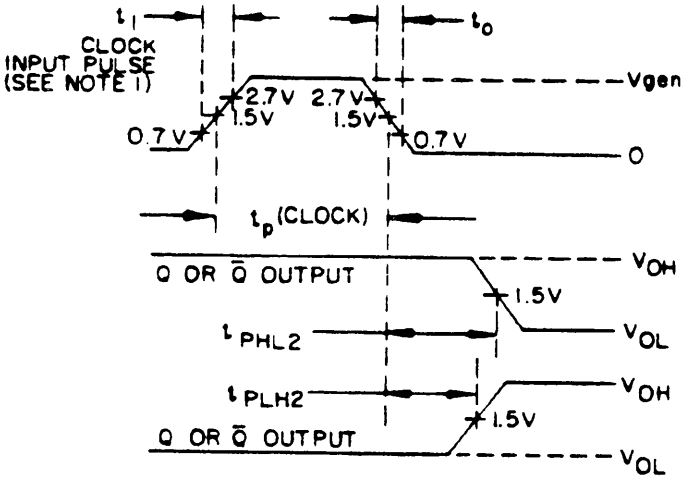
NOTES:
1. Clear inputs dominate regardless of the state of clock or J-K inputs.
2. Clear input pulse characteristics: Vgen = 3 V, t0 = t1 = 10 ns, t_p (clear) = 30 ns, PRR = 1 MHz.
3. All diodes are 1N3064, or equivalent.
4. C_L = 50 pF, minimum (including jig and probe capacitance).
5. R_L = 390Ω ±5%.
6. Clock input pulse characteristics: Vgen = 3 V, t_p (clock) ≥ 25 ns, PRR = 1 MHz.

FIGURE 7. Clear switching test circuit and waveforms for device types 02 and 03.

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TEST CIRCUIT

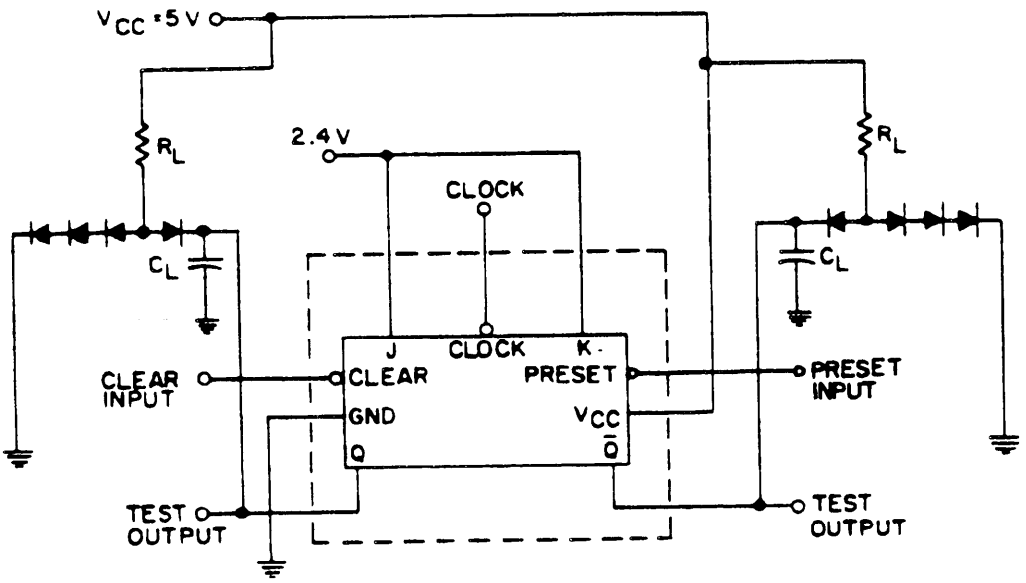


VOLTAGE WAVEFORMS

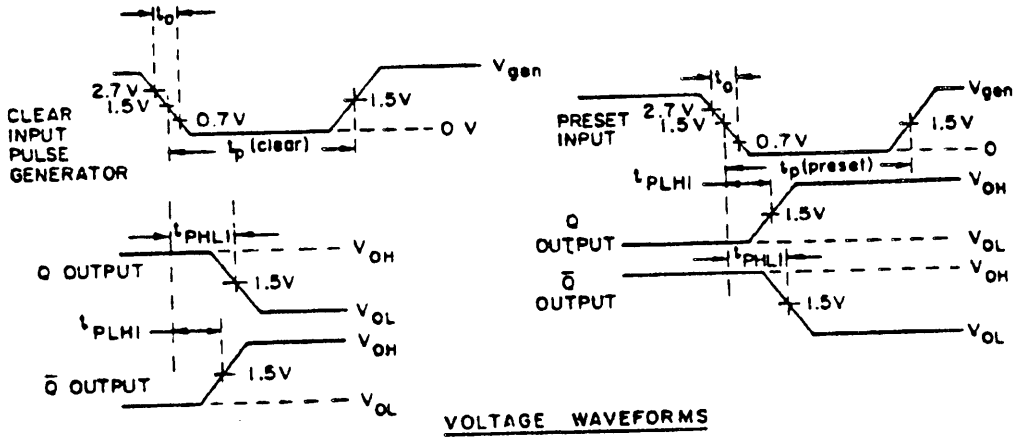
NOTES:
1. Clock input characteristics for t_{PLH}, t_{PHL} (clock to output), V_{gen} = 3 V, $t_1 = t_0 \leq 10$ ns, t_p (clock) = 25 ns, and PRR = 1 MHz. All J and K inputs are at 2.4 V. When testing f_{MAX} the clock input characteristics are V_{gen} = 3 V, $t_1 = t_0 \leq 10$ ns, t_p (clock) = 20 ns, and PRR = 1 MHz.
2. All diodes are 1N3064, or equivalent.
3. C_L = 50 pF minimum (including jig and probe capacitance).
4. R_L = 390 Ω ±5%.

FIGURE 8. Synchronous switching test circuit for device types 02 and 03.

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TEST CIRCUIT

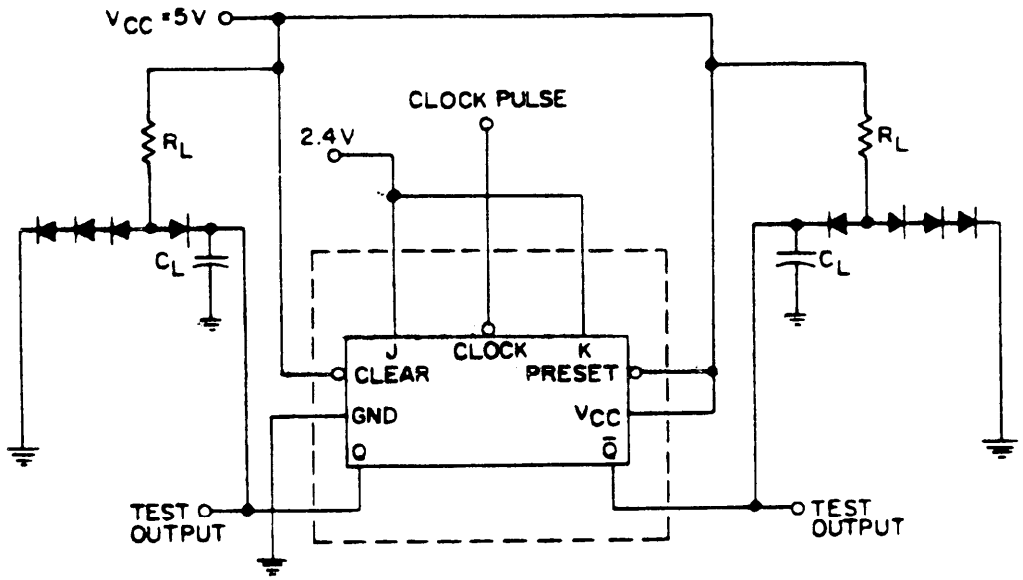


VOLTAGE WAVEFORMS

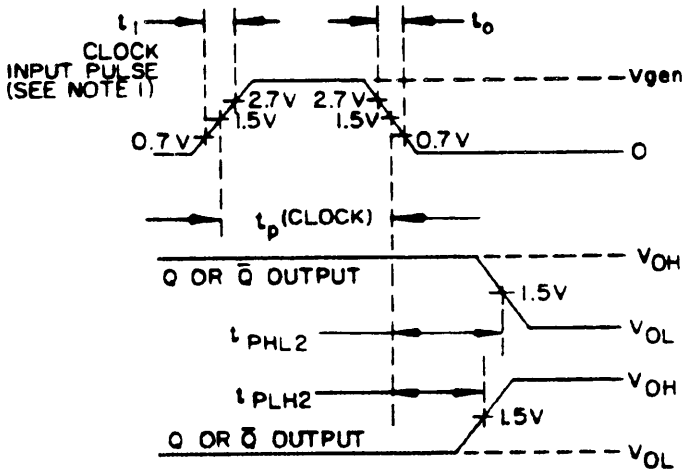
NOTES:
1. Clear or preset inputs dominate regardless of the state of clock or J-K inputs.
2. Clear or preset input pulse characteristics: Vgen = 3 V, $t_0 = t_1 = 10$ ns, t_p (clear) = t_p (preset) = 30 ns, PRR = 1 MHz, and $Z_{out} \approx 50\Omega$.
3. C_L = 50 pF, minimum (including jig and probe capacitance).
4. $R_L = 390\Omega \pm 5\%$.
5. All diodes are 1N3064, or equivalent.
6. When testing clear to output switching, preset input shall have a negative pulse. When testing preset to output switching, clear input shall have a negative pulse (see table III).

FIGURE 9. Clear and preset switching test circuit and waveforms for device type 04.

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TEST CIRCUIT

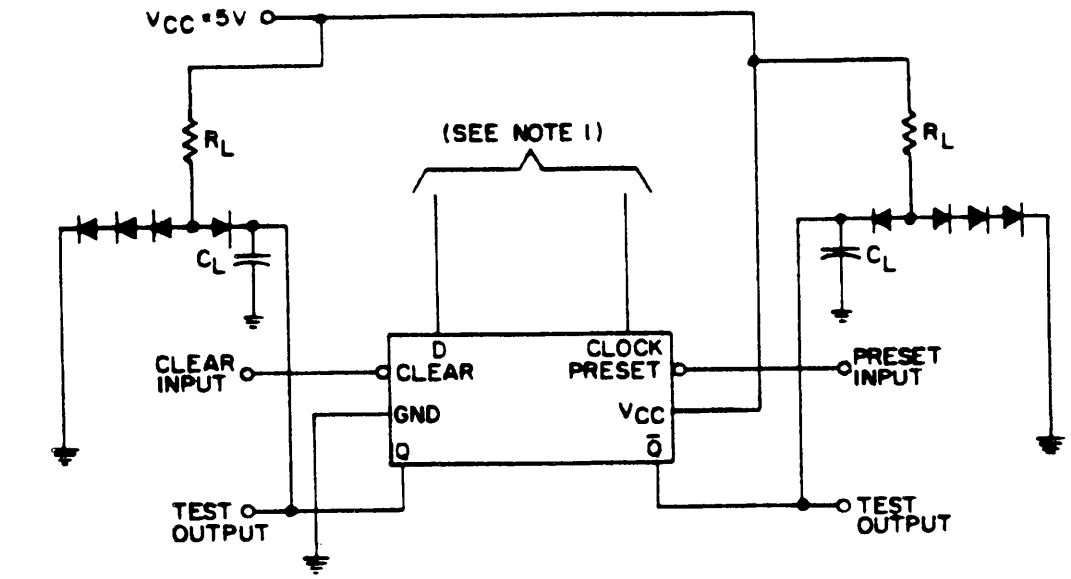


VOLTAGE WAVEFORMS

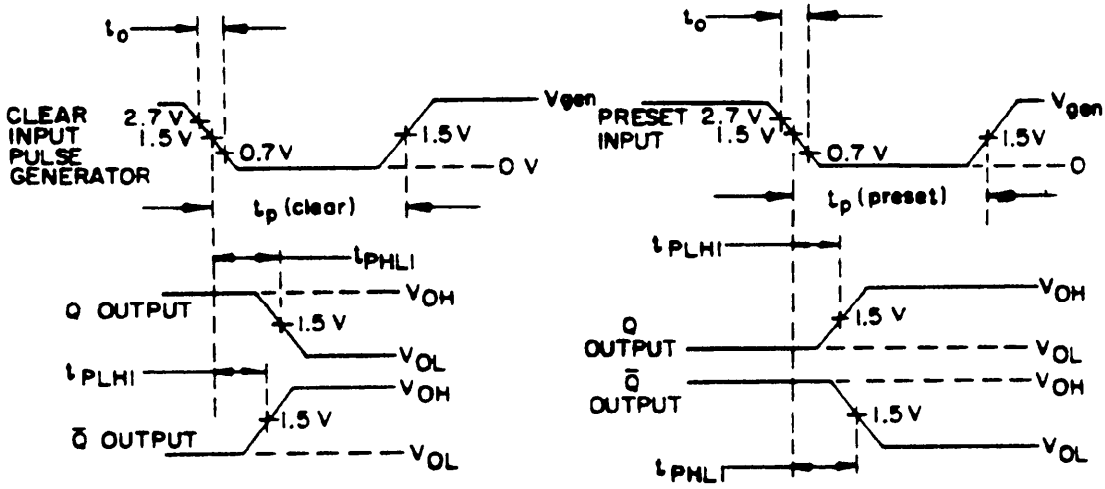
NOTES:
1. Clock input characteristics for t_{PLH}, t_{PHL} (clock to output), Vgen = 3 V, $t_1 = t_0 \leq 10$ ns, t_p (clock) = 25 ns, and PRR = 1 MHz. All J and K inputs are at 2.4 V. When testing f_{MAX} the clock input characteristics are Vgen = 3V, $t_1 = t_0 \leq 10$ ns, t_p (clock) = 20 ns, and PRR = see table III.
2. All diodes are 1N3064, or equivalent.
3. C_L = 50 pF minimum (including jig and probe capacitance).
4. R_L = 390Ω ±5%.

FIGURE 10. Synchronous switching test circuit for device type 04.

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TEST CIRCUIT

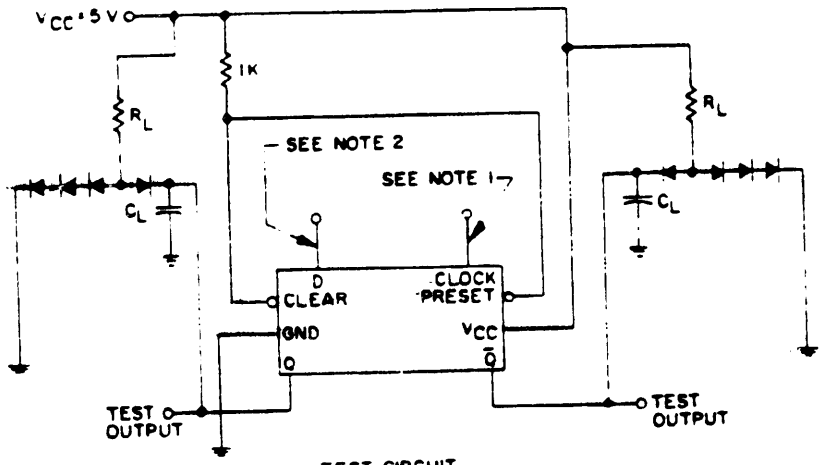


VOLTAGE WAVEFORMS

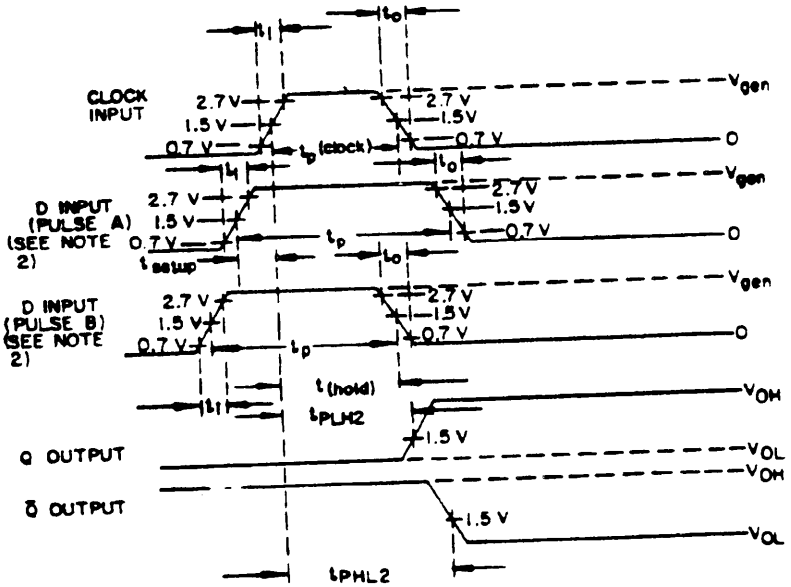
NOTES:
1. Clear and preset inputs dominate regardless of the state of clock or D inputs.
2. All diodes are 1N3064, or equivalent.
3. Clear or preset input pulse characteristics: Vgen = 3 V, $t_0 \leq 7$ ns, t_p (clear) = t_p (preset) = 35 ns, and PRR = 1 MHz.
4. C_L = 50 pF, minimum (including jig and probe capacitance).
5. R_L = 390Ω ±5%.
6. When testing clear to output switching, preset input shall have a negative pulse. When testing preset to output switching, clear input shall have a negative pulse (see table III).

FIGURE 11. Clear and preset switching test circuit and waveforms for device types 05 and 07.

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TEST CIRCUIT



VOLTAGE WAVEFORMS

NOTES:
1. Clock input pulse has the following characteristics: Vgen = 3 V, $t_0 = t_1 \leq 10$ ns, t_p(clock) = 30 ns, and PRR = 1 MHz. When testing f_{MAX}, PRR = see table III.
2. D input (pulse A) has the following characteristics: Vgen = 3 V, $t_0 = t_1 \leq 10$ ns, t_{setup} = 25 ns, t_p = 60 ns, and PRR is 50% of the clock PRR. D input (pulse B) has the following characteristics: Vgen = 3 V, $t_0 = t_1 < 7$ ns, t_{hold} = 6 ns, t_p = 60 ns, and PRR is 50% of the clock PRR.
3. All diodes are 1N3064, or equivalent.
4. C_L = 50 pF, minimum (including jig and probe capacitance).
5. R_L = 390 Ω ±5%.

FIGURE 12. Synchronous switching test circuit (high level data) for device types 05 and 07.

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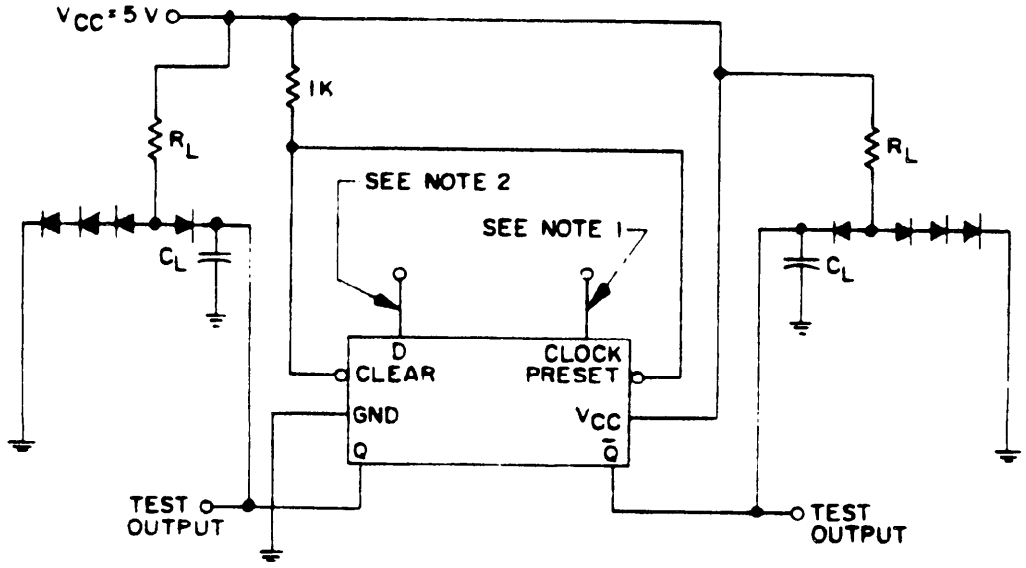
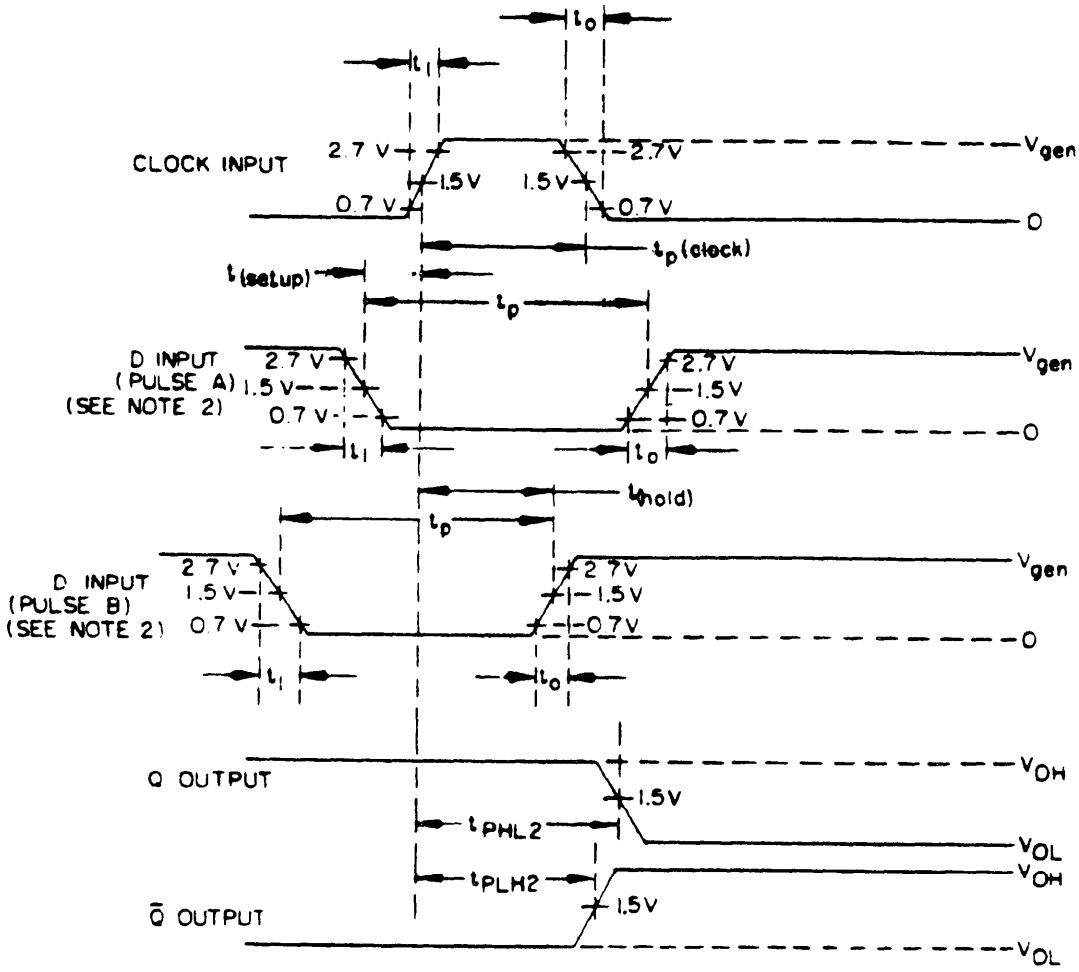


FIGURE 13. Synchronous switching test circuit (low-level data) for device types 05 and 07.

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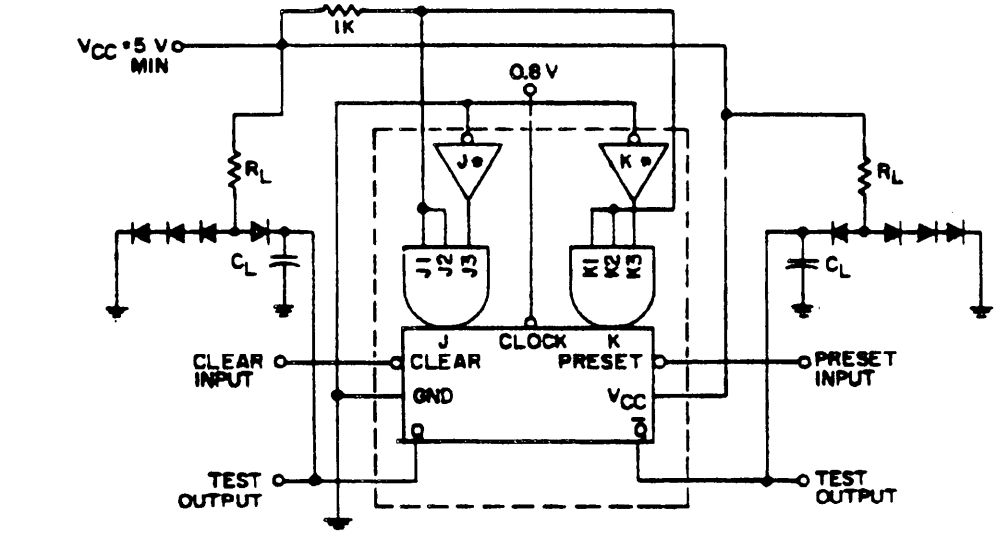


VOLTAGE WAVEFORMS

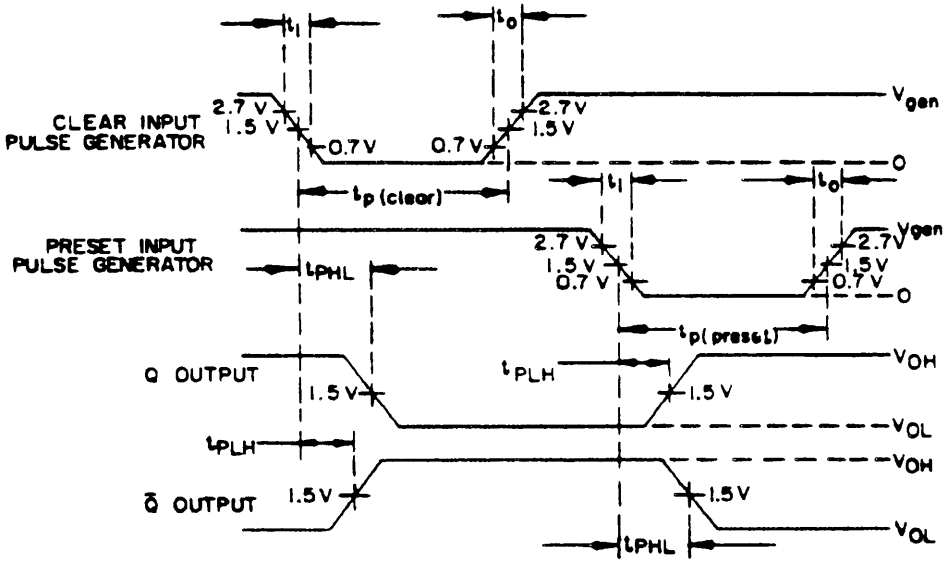
NOTES:
1. Clock input pulse has the following characteristics: V_{gen} = 3 V, $t_0 = t_1 < 10$ ns, t_p (clock) = 30 ns, and PRR = 1 MHz.
2. D input (pulse A) has the following characteristics: V_{gen} = 3 V, $t_0 = t_1 < 10$ ns, t_{setup} = 25 ns, t_p = 60 ns, and PRR is 50% of the clock PRR. D input (pulse B) has the following characteristics: V_{gen} = 3 V, $t_0 = t_1 < 10$ ns, t_{hold} = 6 ns, t_p = 60 ns, and PRR is 50% of the clock PRR.
3. All diodes are 1N3064, or equivalent.
4. C_L = 50 pF, minimum (including jig and probe capacitance).
5. R_L = 390Ω ±5%.

FIGURE 13. Synchronous switching test circuit (low-level data) for device types 05 and 07 - Continued.

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TEST CIRCUIT

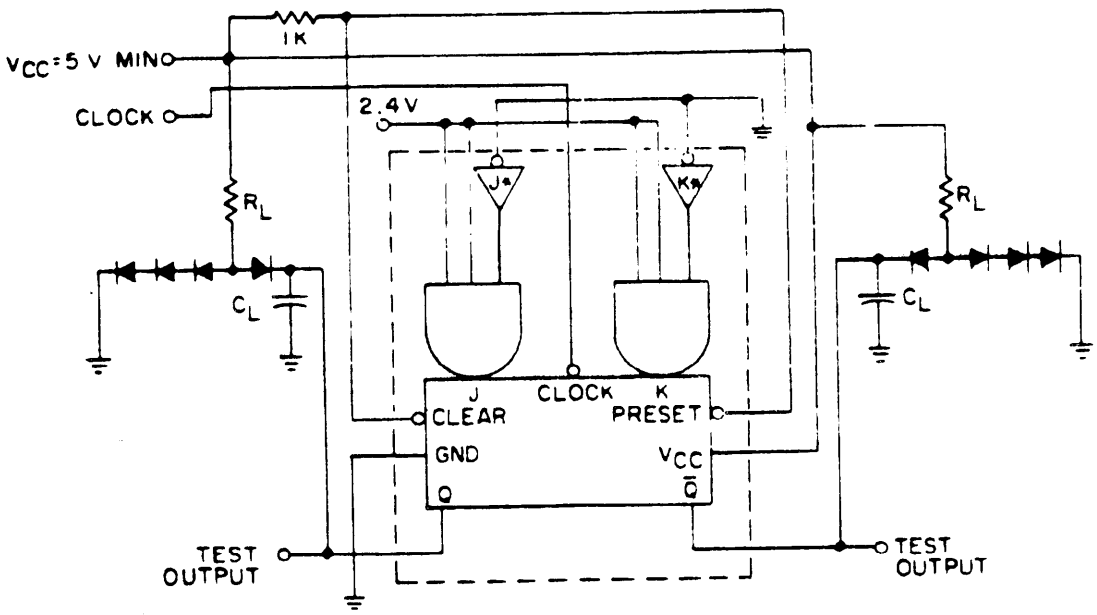


VOLTAGE WAVEFORMS

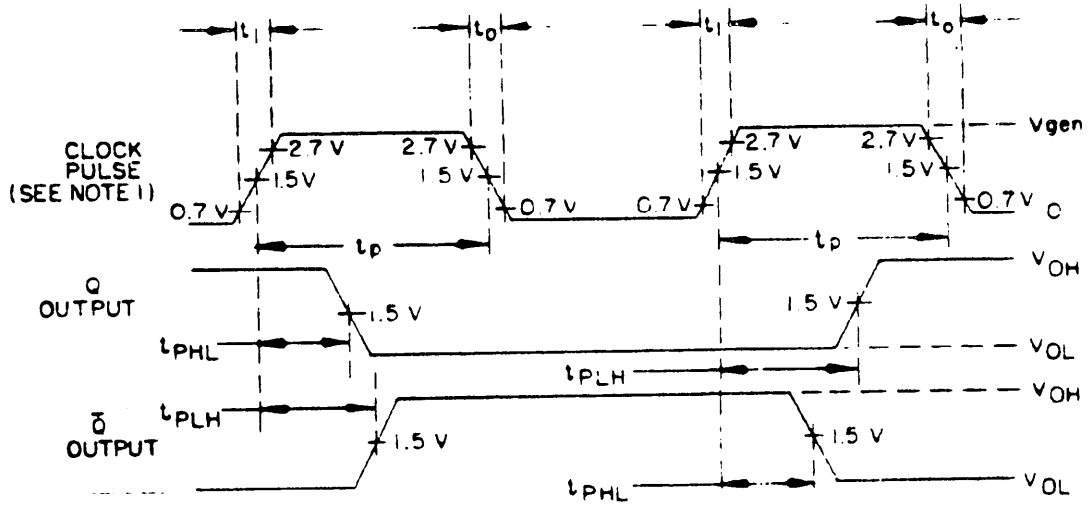
NOTES:
1. Preset or clear function can occur only when clock input is low. Gated inputs are inhibited.
2. All diodes are 1N3064, or equivalent.
3. C_L = 50 pF minimum, including jig and probe capacitance.
4. Clear or preset input pulse characteristics: V_{gen} = 3.0 V, t_0 = 5 ns, $t_1 \leq 10$ ns, t_p = 25 ns.
5. R_L = 390Ω ±5%.

FIGURE 14. Clear and preset switching test circuit and waveforms for device type 06.

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TEST CIRCUIT

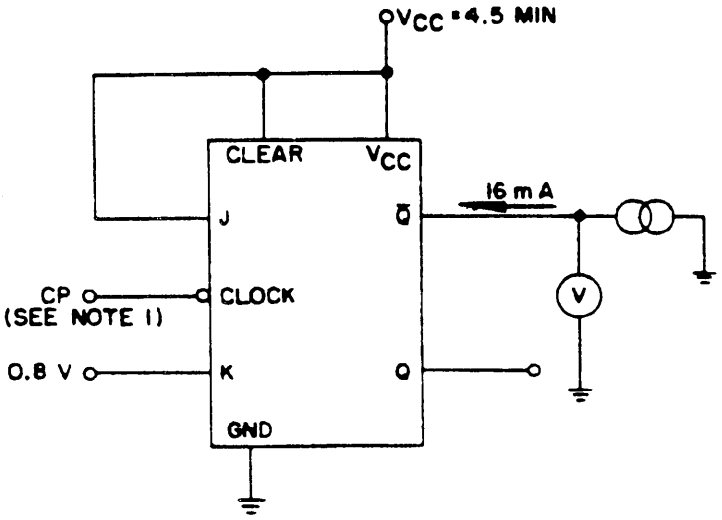


VOLTAGE WAVEFORMS

NOTES:
1. Clock pulse characteristics: Vgen = 3.0 V, $t_1 = t_0 \leq 10$ ns, t_p = 30 ns, and PRR = 1 MHz. When testing f_{MAX}, PRR = see table III.
2. All diodes are 1N3064, or equivalent.
3. C_L = 50 pF minimum, including jig and probe capacitance.
4. R_L = 390Ω ±5%.

FIGURE 15. Synchronous switching test circuit for device type 06.

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NOTES:
1. Apply normal clock pulse, then sink -12 mA on the clock input.
2. The output $\overline{Q}$ is measured after -12 mA is applied to the clock to insure it is still in the low state.

FIGURE 16. Input clamp voltage test circuit for device types 01, 02, 03 and 04 (circuit B).

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TABLE III. Group A inspection for device type 01. 1/

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D / Case C Test No.	1 9 K1	2 12 Clock	3 13 Preset	4 14 V_{CC}	5 2 Clear	6 1 NC	7 3 J1	8 4 J2	9 5 J3	10 6 $\overline{Q}$	11 7 GND	12 8 Q	13 10 K2	14 11 K3	Meas. terminal	Test limits Min	Max	Unit
1 T_A = 25°C	V_{OH}	3006	1	0.8 V	A		4.5 V			2.0 V	2.0 V	2.0 V		GND	-.4 mA	0.8 V	0.8 V	Q	2.4		V
			2	2.0 V	A					0.8 V	0.8 V	0.8 V	-.4 mA			2.0 V	2.0 V	$\overline{Q}$			
			3			0.8 V		2.0 V							-.4 mA			Q			
			4			2.0 V		0.8 V					-.4 mA					$\overline{Q}$			
	V_{OL}	3007	5	2.0 V	A					0.8 V	0.8 V	0.8 V			16 mA	2.0 V	2.0 V	Q		0.4	
			6	0.8 V	A					2.0 V	2.0 V	2.0 V	16 mA			0.8 V	0.8 V	$\overline{Q}$			
			7			0.8 V		2.0 V					16 mA					Q			
			8			2.0 V		0.8 V							16 mA			$\overline{Q}$			
	V_{IC}		9							-12mA								J1		-1.5	
			10								-12 mA							J2			
			11									-12mA						J3			
			12	-12 mA														K1			
			13													-12mA		K2			
			14														-12mA	K3			
			15					-12 mA										Clear			
			16			-12 mA												Preset			
			17		-12 mA													Clock			
			17 CKT B	0.8 V	A*			4.5 V		4.5 V	4.5 V	4.5 V				0.8 V	0.8 V	Clock		-0.5	
	I_{IL1}	3009	18	GND	4.5 V		5.5 V	B		0.4 V	4.5 V	4.5 V				GND	GND	J1	-0.7	-1.6	mA
			19					B		4.5 V	0.4 V	4.5 V				GND	GND	J2			
			20					B		4.5 V	4.5 V	0.4 V				GND	GND	J3			
			21	0.4 V		B				GND	GND	GND				4.5 V	4.5 V	K1			
			22	4.5 V		B				GND	GND	GND				0.4 V	4.5 V	K2			
			23	4.5 V		B				GND	GND	GND				4.5 V	0.4 V	K3			
	I_{IL2}		24 CKT A	4.5 V	0.4 V	B				4.5 V	4.5 V	4.5 V				4.5 V	4.5 V	Clock	-0.7	-1.6	
			24 CKT B, C			B												Clock	-1.4	-3.2	
			25 CKT A					B										Clock	-0.7	-1.6	
			25 CKT B, C					B										Clock	-1.4	-3.2	
	I_{IL3}		26 CKT A			0.4 V												Preset	-0.7	-1.6	
			26 CKT B, C			0.4 V												Preset	-1.4	-3.2	
			27 CKT A					0.4 V										Clear	-0.7	-1.6	
			27 CKT B, C					0.4 V										Clear	-1.4	-3.2	
	I_{IH1}	3010	28		GND		5.5 V	GND		2.4 V	GND	GND						J1	0	40	µA
			29					GND		GND	2.4 V	GND						J2			
			30					GND		GND	GND	2.4 V						J3			
			31	2.4 V		GND										GND	GND	K1			
			32	GND		GND										2.4 V	GND	K2			
			33	GND		GND										GND	2.4 V	K3			
	I_{IH2}		34					GND		5.5 V	GND	GND						J1		100	
			35					GND		GND	5.5 V	GND						J2			
			36					GND		GND	GND	5.5 V						J3			
			37	5.5 V		GND										GND	GND	K1			
			38	GND		GND										5.5 V	GND	K2			
			39	GND		GND										GND	5.5 V	K3			
	I_{IH3}		40	4.5 V	A			2.4 V		GND	GND	GND				4.5 V	4.5 V	Clear		80	
	I_{IH3}		41	GND	A	2.4 V		GND		4.5 V	4.5 V	4.5 V				GND	GND	Preset		80	
	I_{IH4}		42	GND	A	5.5 V		GND		4.5 V	4.5 V	4.5 V				GND	GND	Preset		200	
			43	4.5 V	A			5.5 V		GND	GND	GND				4.5 V	4.5 V	Clear			
			44	GND	5.5 V			GND		GND	GND	GND				GND	GND	Clock			
			45	GND	5.5 V	GND				GND	GND	GND				GND	GND	Clock			

See notes at end of device type 01.

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TABLE III. Group A inspection for device type 01 1/ - Continued.

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D / Case C / Test No.	1 / 9 / K1	2 / 12 / Clock	3 / 13 / Preset	4 / 14 / V_{CC}	5 / 2 / Clear	6 / 1 / NC	7 / 3 / J1	8 / 4 / J2	9 / 5 / J3	10 / 6 / $\overline{Q}$	11 / 7 / GND	12 / 8 / Q	13 / 10 / K2	14 / 11 / K3	Meas. terminal	Test limits Min	Max	Unit
1 T_A = 25°C	I_{IH5}	3010	46 CKT A, C 46 CKT B 47 CKT A, C 47 CKT B	GND	2.4 V 2.4 V 2.4 V 2.4 V	GND GND	5.5 V	GND GND		GND	GND	GND		GND		GND	GND	Clock Clock Clock Clock	-500 -850 -500 -850	-50 -200 -50 -200	µA
	I_{OS} I_{OS}	3011 3011	48 49	4.5 V 4.5 V	GND	GND	5.5 V	GND		4.5 V 4.5 V	4.5 V 4.5 V	4.5 V 4.5 V	GND		GND	4.5 V 4.5 V	4.5 V 4.5 V	Q $\overline{Q}$	-20 -20	-57 -57	mA
	I_{CC} I_{CC}	3005 3005	50 51	GND GND		GND		GND		GND GND	GND GND	GND GND				GND GND	GND GND	V_{CC} V_{CC}		20 20	
2	Same tests, terminal conditions and limits as for subgroup 1, except T_A = 125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_A = -55°C and V_{IC} tests are omitted.																				
7 2/ 4/ T_A = 25°C			52	B	B	A	4.5 V	B	B	B	B	B	H 3/	GND	L 3/	B	B	All output	H or L as shown 3/		
			53	B	B	B		A	B	B	B	B	L		H	B	B				
			54	B	B	A		A	B	B	B	B	L		H	A	A				
			55	B	A	A		A	B	B	B	B	L		H	A	A				
			56	B	B	A		A	B	B	B	B	L		H	A	A				
			57	A	B	A		A	B	B	B	B	L		H	B	A				
			58	A	A	A		A	B	B	B	B	L		H	B	A				
			59	A	B	A		A	B	B	B	B	L		H	B	A				
			60	A	B	A		A	B	B	B	B	L		H	A	B				
			61	A	A	A		A	B	B	B	B	L		H	A	B				
			62	A	B	A		A	B	B	B	B	L		H	A	B				
			63	A	B	A		B	B	B	B	B	H		L	A	B				
			64	B	B	A		A	B	B	A	A	H		L	B	B				
			65	B	A	A		A	B	B	A	A	H		L	B	B				
			66	B	B	A		A	B	B	A	A	H		L	B	B				
			67	B	B	A		A	B	A	B	A	H		L	B	B				
			68	B	A	A		A	B	A	B	A	H		L	B	B				
			69	B	B	A		A	B	A	B	A	H		L	B	B				
			70	B	B	A		A	B	A	A	B	H		L	B	B				
			71	B	A	A		A	B	A	A	B	H		L	B	B				
			72	B	B	A		A	B	A	A	B	H		L	B	B				
			73	A	B	A		A	B	A	A	A	H		L	A	A				
			74	A	A	A		A	B	A	A	A	H		L	A	A				
			75	A	B	A		A	B	A	A	A	L		H	A	A				
			76	A	A	A		A	B	A	A	A	L		H	A	A				
			77	A	B	A		A	B	A	A	A	H		L	A	A				
			78	A	B	A		B	B	A	A	A	H		L	A	A				
			79	A	A	A		B	B	A	A	A	H		L	A	A				
			80	A	B	A		B	B	A	A	A	H		L	A	A				
			81	A	B	B		B	B	A	A	A	H		H	A	A				
			82	A	A	B		B	B	A	A	A	H		H	A	A				
			83	A	B	B		B	B	A	A	A	H		H	A	A				
			84	A	A	B		A	B	A	A	A	L		H	A	A				
			85	A	A	A		A	B	A	A	A	L		H	A	A				
			86	B	A	A		A	B	A	A	A	L		H	A	A				
			87	B	A	A		A	B	B	A	A	L		H	A	A				
			88	B	B	A		A	B	B	A	A	H		L	A	A				
			89	A	A	A		A	B	A	A	A	H		L	A	A				
			90	A	A	A		A	B	B	A	A	H		L	A	A				

See notes at end of device type 01.

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TABLE III. Group A inspection for device type 01 1/ - Continued.

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D / Case C / Test No.	1 / 9 / K1	2 / 12 / Clock	3 / 13 / Preset	4 / 14 / V_{CC}	5 / 2 / Clear	6 / 1 / NC	7 / 3 / J1	8 / 4 / J2	9 / 5 / J3	10 / 6 / $\overline{Q}$	11 / 7 / GND	12 / 8 / Q	13 / 10 / K2	14 / 11 / K3	Meas. terminal	Test Limits: Min	Max	Unit
7 2/ 4/ T_A = 25°C			91	B	A	A	4.5 V	A	B	B	A	A	H	GND	L	A	A	All output	H or L as shown 3/		
			92	B	B	A	4.5 V	A	B	B	A	A	L	GND	H	A	A				
8 2/ 4/	Same tests, terminal conditions and limits as for subgroup 7, except T_A = 125 and -55° C.																				
9 T_A = 25°C	F_{MAX}	(Fig 6)	93	2.4 V	IN	5.0 V	5.0 V	5.0 V		2.4 V	2.4 V	2.4 V	OUT	GND	OUT	2.4 V	2.4 V	$\overline{Q}$ to Q	5		MHz
	F_{MAX}	(Fig 6)	94			5.0 V												Q to $\overline{Q}$	5		MHz
	t_{PLH1}	3003	95		2.4 V	J		IN					OUT		OUT			Clear to Q	5	25	ns
	t_{PLH1}	(Fig 5)	96			IN		J										Preset to Q		25	
	t_{PHL1}		97			J		IN					OUT		OUT			Clear to $\overline{Q}$		40	
	t_{PHL1}		98			IN		J										Preset to $\overline{Q}$		40	
	t_{PLH2}	3003	99		IN	5.0 V		5.0 V					OUT		OUT			Clock to Q		30	ns
	t_{PLH2}	(Fig 6)	100															Clock to $\overline{Q}$		30	
	t_{PHL2}		101										OUT		OUT			Clock to Q		40	
	t_{PHL2}		102															Clock to $\overline{Q}$		40	
10 T_A = 125°C	F_{MAX}	(Fig 6)	103										OUT		OUT			$\overline{Q}$ to Q	5		MHz
	F_{MAX}	(Fig 6)	104															Q to $\overline{Q}$	5		MHz
	t_{PLH1}	3003	105		2.4 V	J		IN					OUT		OUT			Clear to Q	5	39	ns
	t_{PLH1}	(Fig 5)	106			IN		J										Preset to Q		39	
	t_{PHL1}		107			J		IN					OUT		OUT			Clear to $\overline{Q}$		50	
	t_{PHL1}		108			IN		J										Preset to $\overline{Q}$		50	
	t_{PLH2}	(Fig 6)	109		IN	5.0 V		5.0 V					OUT		OUT			Clock to Q		39	ns
	t_{PLH2}		110															Clock to $\overline{Q}$		39	
	t_{PHL2}		111										OUT		OUT			Clock to Q		50	
	t_{PHL2}		112															Clock to $\overline{Q}$		50	
11	Same tests, terminal conditions and limits as subgroup 10, except T_A = -55° C.																				

NOTES:
A = Normal clock pulse.
B = Momentary GND, then 4.5 V.
J = Input pulse tp ≥ 100 ns, PRR = 1 MHz, V_{OL} = 0 V, V_{OH} = 4.5 V
* After clock pulse apply -12 mA to clock pin to insure $\overline{Q}$ is still in the low state (see figure 16).

1/ Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open).
2/ Input voltages shown are: A = 2.0 volts minimum and B = 0.8 volts maximum.
3/ Output voltages shall be either: (a) H = 2.4 V, minimum and L = 0.4 V, maximum when using a high speed checker double comparator; or (b) H ≥ 1.5 V and L < 1.5 V when using a high speed checker single comparator.
4/ Tests shall be performed in sequence.

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TABLE III. Group A inspection for device type 02 1/

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas. terminal	Test limits		
			Case C	1	2	3	4	5	6	7	8	9	10	11	12	13	14				
			Test No.	Clock 1	Clear 1	K1	V_{CC}	Clock 2	Clear 2	J2	$\overline{Q2}$	Q2	K2	GND	Q1	$\overline{Q1}$	J1		Min	Max	Unit
1 T_A = 25°C	V_{OH}	3006	1	A		0.8 V	4.5 V							GND	-.4 mA		2.0 V	Q1	2.4		V
			2	A		2.0 V										-.4mA	0.8 V	$\overline{Q1}$			
			3		0.8 V											-.4mA		J1			
			4					A		2.0 V		-.4mA	0.8 V					Q2			
			5					A		0.8 V	- 4 mA		2.0 V					$\overline{Q2}$			
			6						0.8 V		-.4 mA							K2			
	V_{OL}	3007	7	A		2.0 V									16 mA		0.8 V	Q1		0.4	
			8	A		0.8 V										16 mA	2.0 V	$\overline{Q1}$			
			9		0.8 V										16 mA			J1			
			10					A		0.8 V		16 mA	2.0 V					Q2			
			11					A		2.0 V	16 mA		0.8 V					$\overline{Q2}$			
			12						0.8 V			16 mA						K2			
	V_{IC}		13							-12mA							-12mA	J1		-1.5	
			14															J2			
			15			-12 mA												K1			
			16										-12 mA					K2			
			17		-12mA													Clear 1			
			18	-12 mA														Clock 1			
			18 CKT B	A*	4.5 V	0.8 V											4.5 V	Clock 1		-0.5	
			19						-12 mA									Clear 2		-1.5	
			20					-12 mA										Clock 2		-1.5	
			20 CKT B					A*	4.5 V	4.5 V			0.8 V					Clock 2		-0.5	
	I_{IL1}	3009	21	4.5 V	B		5.5 V										0.4 V	J1	-.7	-1.6	mA
			22	4.5 V	4.5 V	0.4 V												K1			
			23					4.5 V	B	0.4 V								J2			
			24					4.5 V	4.5 V				0.4 V					K2			
	I_{IL2}		25	0.4 V	B	4.5 V											4.5 V	Clock 1	-1.4	-3.2	
	I_{IL2}		26					0.4 V	B	4.5 V			4.5 V					Clock 2	-1.4	-3.2	
	I_{IL3}		27 CKT A, C	4.5 V	0.4 V	4.5 V											4.5 V	Clear 1	-0.7	-1.6	
			27 CKT B	4.5 V	0.4 V	4.5 V											4.5 V	Clear 1	-1.4	-3.2	
			28 CKT A, C					4.5 V	0.4 V	4.5 V			4.5 V					Clear 2	-0.7	-1.6	
			28 CKT B					4.5 V	0.4 V	4.5 V			4.5 V					Clear 2	-1.4	-3.2	
	I_{IH1}	3010	29	GND	GND		5.5 V										2.4 V	J1	0	40	µA
			30	GND	B	2.4 V												K1			
			31					GND	GND	2.4 V								J2			
			32					GND	B				2.4 V					K2			
	I_{IH2}		33	GND	GND												5.5 V	J1		100	
			34	GND	B	5.5 V												K1			
			35					GND	GND	5.5 V								J2			
			36					GND	B				5.5 V					K2			
	I_{IH3}		37	GND	E												GND	Clear 1		80	
	I_{IH3}		38					GND	E	GND								Clear 2		80	
	I_{IH4}		39	5.5 V	5.5 V	GND											GND	Clock 1		200	
			40	GND	F												GND	Clear 1			
			41					5.5 V	5.5 V	GND			GND					Clock 2			
			42					GND	F	GND								Clear 2			
	I_{IH5}		43 CKT A, C	2.4 V	2.4 V	GND											GND	Clock 1	-500	-50	
			43 CKT B	2.4 V	2.4 V	GND											GND	Clock 1	-850	-200	
			44 CKT A, C					2.4 V	2.4 V	GND			GND					Clock 2	-500	-50	
			44 CKT B					2.4 V	2.4 V	GND			GND					Clock 2	-850	-200	

See notes at end of device type 02.

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TABLE III. Group A inspection for device type 02 1/ - Continued.

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D / Case C / Test No.	1 / 1 / Clock 1	2 / 2 / Clear 1	3 / 3 / K1	4 / 4 / V_{CC}	5 / 5 / Clock 2	6 / 6 / Clear 2	7 / 7 / J2	8 / 8 / $\overline{Q2}$	9 / 9 / Q2	10 / 10 / K2	11 / 11 / GND	12 / 12 / Q1	13 / 13 / $\overline{Q1}$	14 / 14 / J1	Meas. terminal	Test limits Min	Max	Unit
1 T_A = 25°C	I_{OS}	3011 3011** 3011** 3011	45 46 47 48	2.4 V GND	GND 2.4 V	2.4 V 2.4 V	5.5 V	 GND 2.4 V	 2.4 V GND	 2.4 V 2.4 V	 GND GND	GND	 2.4 V 2.4 V	GND	GND	GND GND	2.4 V 2.4 V	$\overline{Q1}$ Q1 $\overline{Q2}$ Q2	-20	-57	mA
	I_{CC}	3005	49	D	4.5 V	GND		D	4.5 V	4.5 V			GND				4.5 V	V_{CC}		40	
2	Same tests, terminal conditions and limits as for subgroup 1, except T_A = 125° C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_A = -55° C and V_{IC} tests are omitted.																				
7 2/ 4/ T_A = 25°C			50	B	B	B	4.5 V	B	B	A	H 3/	L 3/	B	GND	L 3/	H 3/	A	All output	H or L as shown 3/		
			51	A	B	B		A	B	A	H	L	B		L	H	A				
			52	B	B	B		B	B	A	H	L	B		L	H	A				
			53	B	B	A		B	B	A	H	L	A		L	H	A				
			54	A	B	A		A	B	A	H	L	A		L	H	A				
			55	B	B	A		B	B	A	H	L	A		L	H	A				
			56	B	A	A		B	A	A	H	L	A		L	H	A				
			57	A	A	A		A	A	A	H	L	A		L	H	A				
			58	B	A	A		B	A	A	L	H	A		H	L	A				
			59	A	A	A		A	A	A	L	H	A		H	L	A				
			60	B	A	A		B	A	A	H	L	A		L	H	A				
			61	A	A	A		A	A	A	H	L	A		L	H	A				
			62	B	A	A		B	A	A	L	H	A		H	L	A				
			63	B	A	B		B	A	B	L	H	B		H	L	B				
			64	A	A	B		A	A	B	L	H	B		H	L	B				
			65	B	A	B		B	A	B	L	H	B		H	L	B				
			66	B	B	B		B	B	B	H	L	B		L	H	B				
			67	B	A	B		B	A	B	H	L	B		L	H	B				
			68	A	A	B		A	A	B	H	L	B		L	H	B				
			69	B	A	B		B	A	B	H	L	B		L	H	B				
			70	B	A	B		B	A	A	H	L	B		L	H	A				
			71	A	A	B		A	A	A	H	L	B		L	H	A				
			72	B	A	B		B	A	A	L	H	B		H	L	A				
			73	B	A	A		B	A	B	L	H	A		H	L	B				
			74	A	A	A		A	A	B	L	H	A		H	L	B				
			75	B	A	A		B	A	B	H	L	A		L	H	B				
			76	A	B	A		A	B	A	H	L	A		L	H	A				
			77	A	A	A		A	A	A	H	L	A		L	H	A				
			78	A	A	A		A	A	B	H	L	A		L	H	A				
			79	A	A	B		A	A	B	H	L	B		L	H	B				
			80	B	A	B		B	A	B	L	H	B		H	L	B				
			81	A	A	A		A	A	A	L	H	A		H	L	A				
			82	A	A	B		A	A	A	L	H	B		H	L	A				
			83	A	A	B		A	A	B	L	H	B		H	L	B				
			84	B	A	B		B	A	B	H	L	B		L	H	B				
			85	A	A	B		A	A	A	H	L	B		L	H	A				
			86	B	A	B		B	A	A	L	H	B		H	L	A				
			87	A	A	B		A	A	A	L	H	B		H	L	A				
			88	A	B	B		A	B	A	H	L	B		L	H	A				
8 2/ 4/	Same tests, terminal conditions and limits as for subgroup 7, except T_A = 125 and -55° C																				

See notes at end of device type 02.

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Subgroup	Symbol	MIL-STD-883 method	Case A, B, D / Case C Test No.	1 Clock 1	2 Clear 1	3 K1	4 V_{CC}	5 Clock 2	6 Clear 2	7 J2	8 $\overline{Q2}$	9 Q2	10 K2	11 GND	12 Q1	13 $\overline{Q1}$	14 J1	Meas. terminal	Test limits Min	Max	Unit
9 T_A = 25°C	F_{MAX}	(Fig 8)	89	IN	5.0 V	2.4 V	5.0 V							GND	OUT		2.4 V	Q1	5		MHz
			90	IN	5.0 V	2.4 V										OUT	2.4 V	Q1			
			91					IN	5.0 V	2.4 V		OUT	2.4 V					$\overline{Q2}$			
			92					IN	5.0 V	2.4 V	OUT		2.4 V					$\overline{Q2}$			
	t_{PLH1}	3003 (Fig 7)	93	IN	IN	GND										OUT	2.4 V	Clear 1 to $\overline{Q1}$	5	25	ns
	t_{PLH1}		94					IN	IN	2.4 V	OUT		GND					Clear 2 to $\overline{Q2}$		25	
	t_{PHL1}		95	IN	IN	GND									OUT		2.4 V	Clear 1 to Q1		40	
	t_{PHL1}		96					IN	IN	2.4 V		OUT	GND					Clear 2 to $\overline{Q2}$		40	
	t_{PLH2}	3003 (Fig 8)	97	IN	J	2.4 V									OUT		2.4 V	Clock 1 to Q1		30	
			98	IN	J	2.4 V										OUT	2.4 V	Clock 1 to $\overline{Q1}$			
			99					IN	J	2.4 V		OUT	2.4 V					Clock 2 to Q2			
			100					IN	J	2.4 V	OUT		2.4 V					Clock 2 to $\overline{Q2}$			
	t_{PHL2}		101	IN	J	2.4 V									OUT		2.4 V	Clock 1 to Q1		40	
			102	IN	J	2.4 V										OUT	2.4 V	Clock 1 to $\overline{Q1}$			
			103					IN	J	2.4 V		OUT	2.4 V					Clock 2 to Q2			
			104					IN	J	2.4 V	OUT		2.4 V					Clock 2 to $\overline{Q2}$			
10 T_A = 125°C	F_{MAX}	(Fig 8)	105	IN	5.0 V	2.4 V									OUT		2.4 V	Q1	5		MHz
			106	IN	5.0 V	2.4 V										OUT	2.4 V	Q1			
			107					IN	5.0 V	2.4 V		OUT	2.4 V					$\overline{Q2}$			
			108					IN	5.0 V	2.4 V	OUT		2.4 V					$\overline{Q2}$			
	t_{PLH1}	3003 (Fig 7)	109	IN	IN	GND										OUT	2.4 V	Clear 1 to $\overline{Q1}$	5	39	ns
	t_{PLH1}		110					IN	IN	2.4 V	OUT		GND					Clear 2 to $\overline{Q2}$		39	
	t_{PHL1}		111	IN	IN	GND									OUT		2.4 V	Clear 1 to Q1		50	
	t_{PHL1}		112					IN	IN	2.4 V		OUT	GND					Clear 2 to $\overline{Q2}$		50	
	t_{PLH2}	3003 (Fig 8)	113	IN	J	2.4 V									OUT		2.4 V	Clock 1 to Q1		39	
			114	IN	J	2.4 V										OUT	2.4 V	Clock 1 to $\overline{Q1}$			
			115					IN	J	2.4 V		OUT	2.4 V					Clock 2 to Q2			
			116					IN	J	2.4 V	OUT		2.4 V					Clock 2 to $\overline{Q2}$			

See notes at end of device type 02.

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TABLE III. Group A inspection for device type 02 1/ - Continued.

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D / Case C / Test No.	1 / 1 / Clock 1	2 / 2 / Clear 1	3 / 3 / K1	4 / 4 / V_{CC}	5 / 5 / Clock 2	6 / 6 / Clear 2	7 / 7 / J2	8 / 8 / $\overline{Q}2$	9 / 9 / Q2	10 / 10 / K2	11 / 11 / GND	12 / 12 / Q1	13 / 13 / $\overline{Q}1$	14 / 14 / J1	Meas. terminal	Test limits: Min	Max	Unit
10 T_A = 125°C	t_{PHL2}	3003	117	IN	J	2.4 V	5.0 V							GND	OUT		2.4 V	Clock 1 to Q1	5	50	ns
			118	IN	J	2.4 V										OUT	2.4 V	Clock 1 to $\overline{Q}1$			
			119					IN	J	2.4 V		OUT	2.4 V					Clock 2 to Q2			
			120					IN	J	2.4 V	OUT		2.4 V					Clock 2 to $\overline{Q}2$			
11	Same tests, terminal conditions and limits as subgroup 10, except T_A = -55°C.																				

NOTES:
- A = Normal clock pulse.
- B = Momentary GND, then 4.5 V.
- C = Momentary GND, then open.
- D = Momentary 4.5 V, then GND.
- E = Momentary ground, then 2.4 volts.
- F = Momentary ground, then 5.5 volts.
- J = Input pulse $t_p \geq 100$ ns, PRR = 1 MHz, V_{OL} = 0 V, V_{OH} = 4.5 V.
- * After clock pulse apply -12 mA to clock pin to insure $\overline{Q}$ is still in the low state (see figure 16).
- ** = Test time limit 100 ms.

1/ Terminal conditions (pins not designated may be H $\geq$ 2.0 V, or L $\leq$ 0.8 V, or open).
2/ Input voltages shown are: A = 2.0 volts minimum and B = 0.8 volts maximum.
3/ Output voltages shall be either: (a) H = 2.4 V, minimum and L = 0.4 V, maximum when using a high speed checker double comparator; or (b) H $\geq$ 1.5 V and L < 1.5 V when using a high speed checker single comparator.
4/ Tests shall be performed in sequence.

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TABLE III. Group A inspection for device type 03 1/

Subgroup	Symbol	MIL-STD-883 method	Case C Test No.	1 J1	2 $\overline{Q}$1	3 Q1	4 K1	5 Q2	6 $\overline{Q}$2	7 GND	8 J2	9 Clock 2	10 Clear 2	11 K2	12 Clock 1	13 Clear 1	14 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
1 T_A = 25°C	V_{OH}	3006	1	2.0 V			0.8 V			GND					A		4.5 V	Q1	2.4		V
			2	0.8 V	-.4mA	-.4mA	2.0 V								A			$\overline{Q}$1			
			3		-.4mA											0.8 V		Q1			
			4					-.4 mA			2.0 V	A		0.8 V				$\overline{Q}$2			
			5						-.4 mA		0.8 V	A		2.0 V				Q2			
			6						-.4 mA				0.8 V					$\overline{Q}$2			
	V_{OL}	3007	7	0.8 V		16 mA	2.0 V								A			$\overline{Q}$1		0.4	
			8	2.0 V	16 mA		0.8 V								A			Q1			
			9			16 mA										0.8 V		Q1			
			10					16 mA			0.8 V	A		2.0 V				Q2			
			11						16 mA		2.0 V	A		0.8 V				$\overline{Q}$2			
			12					16 mA					0.8 V					Q2			
	V_{IC}		13	-12 mA			-12mA				-12 mA			-12mA				J1		-1.5	
			14															K1			
			15															J2			
			16															K2			
			17											-12mA		-12mA		Clear 1			
			18												-12 mA			Clock 1			
			18 CKT B	4.5 V			0.8 V								A*	4.5 V		Clock 1		-0.5	
			19										-12 mA					Clear 2		-1.5	
			20									-12mA						Clock 2		-1.5	
			20 CKT B								4.5 V	A*	4.5 V	0.8 V				Clock 2		-0.5	
	I_{IL1}	3009	21	0.4 V											4.5 V	B	5.5 V	J1	-0.7	-1.6	mA
			22				0.4 V								4.5 V	4.5 V		K1			
			23								0.4 V	4.5 V	B					J2			
			24									4.5 V	4.5 V	0.4 V				K2			
	I_{IL2}		25	4.5 V			4.5 V								0.4 V	B		Clock 1	-1.4	-3.2	
	I_{IL2}		26								4.5 V	0.4 V	B	4.5 V				Clock 2	-1.4	-3.2	
	I_{IL3}		27 CKT A, C	4.5 V											4.5 V	0.4 V		Clear 1	-0.7	-1.6	
			27 CKT B	4.5 V											4.5 V	0.4 V		Clear 1	-1.4	-3.2	
			28 CKT A, C								4.5 V	4.5 V	0.4 V					Clear 2	-0.7	-1.6	
			28 CKT B								4.5 V	4.5 V	0.4 V					Clear 2	-1.4	-3.2	
	I_{IH1}	3010	29	2.4 V			2.4 V								GND	GND		J1	0	40	μA
			30								2.4 V	GND	GND		GND	B		K1			
			31									GND	B	2.4 V				J2			
			32															K2			
	I_{IH2}		33	5.5 V			5.5 V								GND	GND		J1		100	
			34								5.5 V	GND	GND		GND	B		K1			
			35									GND	B	5.5 V				J2			
			36															K2			
	I_{IH3}		37	GND											GND	E		Clear 1		80	
	I_{IH3}		38								GND	GND	E					Clear 2		80	
	I_{IH4}		39	GND			GND								5.5 V	5.5 V		Clock 1		200	
			40	GND											GND	F		Clear 1			
			41								GND	5.5 V	5.5 V	GND				Clock 2			
			42								GND	GND	F					Clear 2			
	I_{IH5}		43 CKT A, C	GND			GND								2.4 V	2.4 V		Clock 1	-500	-50	
			43 CKT B	GND			GND								2.4 V	2.4 V		Clock 1	-850	-200	
			44 CKT A, C								GND	2.4 V	2.4 V	GND				Clock 2	-500	-50	
			44 CKT B								GND	2.4 V	2.4 V	GND				Clock 2	-850	80	

See notes at end of device type 03.

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TABLE III. Group A inspection for device type 03 1/ - Continued.

Subgroup	Symbol	MIL-STD-883 method	Case C Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas. terminal	Test limits Min	Max	Unit
				J1	$\overline{Q}$1	Q1	K1	Q2	$\overline{Q}$2	GND	J2	Clock 2	Clear 2	K2	Clock 1	Clear 1	V_{CC}				
1 T_A = 25°C	I_{OS}	3011	45	2.4 V	GND		2.4 V								2.4 V	GND		$\overline{Q}$1	-20	-57	mA
		3011*	46	2.4 V	GND	GND	2.4 V	GND	GND		2.4 V	GND	2.4 V	2.4 V	GND	2.4 V		Q1			
		3011*	47						GND		2.4 V	2.4 V	GND	2.4 V				Q2			
		3011	48															$\overline{Q}$2			
	I_{CC}	3005	49	4.5 V			GND				4.5 V	D	4.5 V	GND	D	4.5 V		V_{CC}		40	
2	Same tests, terminal conditions and limits as subgroup 1, except T_A = 125° C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as subgroup 1, except T_A = -55° C and V_{IC} tests are omitted.																				
7 2/ 4/ T_A = 25°C			50	A	H 3/	L 3/	B	L 3/	H 3/	GND	A	B	B	B	B	B	4.5 V	All output	H or L as shown 3/		
			51	A	H	L	B	L	H		A	A	B	B	A	B					
			52	A	H	L	B	L	H		A	B	B	B	B	B					
			53	A	H	L	A	L	H		A	B	B	A	B	B					
			54	A	H	L	A	L	H		A	A	B	A	A	B					
			55	A	H	L	A	L	H		A	B	B	A	B	B					
			56	A	H	L	A	L	H		A	B	A	A	B	A					
			57	A	H	L	A	L	H		A	A	A	A	A	A					
			58	A	L	H	A	H	L		A	B	A	A	B	A					
			59	A	L	H	A	H	L		A	A	A	A	A	A					
			60	A	H	L	A	L	H		A	B	A	A	B	A					
			61	A	H	L	A	L	H		A	A	A	A	A	A					
			62	A	L	H	A	H	L		A	B	A	A	B	A					
			63	B	L	H	B	H	L		B	B	A	B	B	A					
			64	B	L	H	B	H	L		B	A	A	B	A	A					
			65	B	L	H	B	H	L		B	B	A	B	B	A					
			66	B	H	L	B	L	H		B	B	B	B	B	B					
			67	B	H	L	B	L	H		B	B	A	B	B	A					
			68	B	H	L	B	L	H		B	A	A	B	A	A					
			69	B	H	L	B	L	H		B	B	A	B	B	A					
			70	A	H	L	B	L	H		A	B	A	B	B	A					
			71	A	H	L	B	L	H		A	A	A	B	A	A					
			72	A	L	H	B	H	L		A	B	A	B	B	A					
			73	B	L	H	A	H	L		B	B	A	A	B	A					
			74	B	L	H	A	H	L		B	A	A	A	A	A					
			75	B	H	L	A	L	H		B	B	A	A	B	A					
			76	A	H	L	A	L	H		A	A	B	A	A	B					
			77	A	H	L	A	L	H		A	A	A	A	A	A					
			78	B	H	L	A	L	H		B	A	A	A	A	A					
			79	B	H	L	B	L	H		B	A	A	B	A	A					
			80	B	L	H	B	H	L		B	B	A	B	B	A					
			81	A	L	H	A	H	L		A	A	A	A	A	A					
			82	A	L	H	B	H	L		A	A	A	B	A	A					
			83	B	L	H	B	H	L		B	A	A	B	A	A					
			84	B	H	L	B	L	H		B	B	A	B	B	A					
			85	A	H	L	B	L	H		A	A	A	B	A	A					
			86	A	L	H	B	H	L		A	B	A	B	B	A					
			87	A	L	H	B	H	L		A	A	A	B	A	A					
			88	A	H	L	B	L	H		A	A	B	B	A	B					
8 2/ 4/	Same tests, terminal conditions and limits as for subgroup 7, except T_A = 125 and -55° C.																				

See notes at end of device type 03.

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TABLE III. Group A inspection for device type 03 1/ - Continued.

Subgroup	Symbol	MIL-STD-883 method	Case C Test No.	1 J1	2 $\overline{Q}$1	3 Q1	4 K1	5 Q2	6 $\overline{Q}$2	7 GND	8 J2	9 Clock 2	10 Clear 2	11 K2	12 Clock 1	13 Clear 1	14 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
9 T_A = 25°C	F_{MAX}	(Fig 8)	89	2.4 V		OUT	2.4 V			GND					IN	5.0 V	5.0 V	Q1	5		MHz
			90	2.4 V	OUT		2.4 V								IN	5.0 V		$\overline{Q}$1			
			91					OUT			2.4 V	IN	5.0 V	2.4 V				Q2			
			92						OUT		2.4 V	IN	5.0 V	2.4 V				$\overline{Q}$2			
	t_{PLH}	3003	93	2.4 V	OUT		2.4 V								A	IN		Clr 1 to Q1	5	25	ns
	t_{PLH}	(Fig 7)	94						OUT		2.4 V	A	IN	2.4 V				Clr 2 to $\overline{Q}$2		25	
	t_{PHL}		95	2.4 V		OUT	2.4 V								A	IN		Clr 1 to Q1		40	
	t_{PHL}		96					OUT			2.4 V	A	IN	2.4 V				Clr 2 to $\overline{Q}$2		40	
	t_{PLH}	3003	97	2.4 V		OUT	2.4 V								IN	5.0 V		Clk 1 to Q1	5	30	ns
		(Fig 8)	98	2.4 V	OUT		2.4 V								IN	5.0 V		Clk 1 to $\overline{Q}$1			
			99					OUT			2.4 V	IN	5.0 V	2.4 V				Clk 2 to Q2			
			100						OUT		2.4 V	IN	5.0 V	2.4 V				Clk 2 to $\overline{Q}$2			
	t_{PHL}		101	2.4 V		OUT	2.4 V								IN	5.0 V		Clk 1 to Q1		40	
			102	2.4 V	OUT		2.4 V								IN	5.0 V		Clk 1 to $\overline{Q}$1			
			103					OUT			2.4 V	IN	5.0 V	2.4 V				Clk 2 to Q2			
			104						OUT		2.4 V	IN	5.0 V	2.4 V				Clk 2 to $\overline{Q}$2			
10 T_A = 125°C	F_{MAX}	(Fig 8)	105	2.4 V		OUT	2.4 V								IN	5.0 V		Q1	5		MHz
			106	2.4 V	OUT		2.4 V								IN	5.0 V		$\overline{Q}$1			
			107					OUT			2.4 V	IN	5.0 V	2.4 V				Q2			
			108						OUT		2.4 V	IN	5.0 V	2.4 V				$\overline{Q}$2			
	t_{PLH}	3003	109	2.4 V	OUT		2.4 V								A	IN		Clr 1 to Q1	5	39	ns
	t_{PLH}	(Fig 7)	110						OUT		2.4 V	A	IN	2.4 V				Clr 2 to $\overline{Q}$2		39	
	t_{PHL}		111	2.4 V		OUT	2.4 V								A	IN		Clr 1 to Q1		50	
	t_{PHL}		112					OUT			2.4 V	A	IN	2.4 V				Clr 2 to $\overline{Q}$2		50	
	t_{PLH}	3003	113	2.4 V		OUT	2.4 V								IN	5.0 V		Clk 1 to Q1	5	39	ns
		(Fig 8)	114	2.4 V	OUT		2.4 V								IN	5.0 V		Clk 1 to $\overline{Q}$1			
			115					OUT			2.4 V	IN	5.0 V	2.4 V				Clk 2 to Q2			
			116						OUT		2.4 V	IN	5.0 V	2.4 V				Clk 2 to $\overline{Q}$2			
	t_{PHL}		117	2.4 V		OUT	2.4 V								IN	5.0 V		Clk 1 to Q1		50	
			118	2.4 V	OUT		2.4 V								IN	5.0 V		Clk 1 to $\overline{Q}$1			
			119					OUT			2.4 V	IN	5.0 V	2.4 V				Clk 2 to Q2			
			120						OUT		2.4 V	IN	5.0 V	2.4 V				Clk 2 to $\overline{Q}$2			
11	Same tests, terminal conditions and limits as subgroup 10, except T_A = -55° C.																				

NOTES:
A = Normal clock pulse.
B = Momentary GND, then 4.5 V.
C = Momentary GND, then open.
D = Momentary 4.5 V, then GND.
E = Momentary ground, then 2.4 V.
F = Momentary ground, then 5.5 V.
* After clock pulse apply -12 mA to clock pin to insure $\overline{Q}$ is still in the low state (see figure 16).
** Test time limit $\leq$ 100 ns.

1/ Terminal conditions (pins not designated may be H $\geq$ 2.0 V, or L $\leq$ 0.8 V, or open).
2/ Input voltages shown are: A = 2.0 volts minimum and B = 0.8 volts maximum.
3/ Output voltages shall be either: (a) H = 2.4 V, minimum and L = 0.4 V, maximum when using a high speed checker double comparator; or (b) H $\geq$ 1.5 V and L < 1.5 V when using a high speed checker single comparator.
4/ Tests shall be performed in sequence.

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TABLE III. Group A inspection for device type 04 1/

Subgroup	Symbol	MIL-STD-883 method	Case E & F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test Limits		
			Test No.	Clock 1	Preset 1	Clear 1	J1	V_{CC}	Clock 2	Preset 2	Clear 2	J2	$\overline{Q}$2	Q1	K2	GND	$\overline{Q}$1	Q1	K1		Min	Max	Unit
1 T_A = 25°C	V_{OH}	3006	1	A			2.0 V	4.5 V								GND		-.4mA	0.8 V	Q1	2.4		V
			2	A			0.8 V										-.4 mA		2.0 V	$\overline{Q}$1			
			3		2.0 V	0.8 V											-.4 mA			J1			
			4		0.8 V	2.0 V												-.4mA		K1			
			5						A			2.0 V		-.4 mA	0.8 V					J2			
			6						A			0.8 V	-.4 mA		2.0 V					K2			
			7							2.0 V	0.8 V		-.4 mA							Clock 1			
			8							0.8 V	2.0 V			-.4 mA						Preset 1			
	V_{OL}	3007	9	A			0.8 V											16 mA	2.0 V	Clear 1		0.4	
			10	A			2.0 V										16 mA		0.8 V	Clock 2			
			11		0.8 V	2.0 V											16 mA			Preset 2			
			12		2.0 V	0.8 V												16 mA		Clear 2			
			13						A			0.8 V		16 mA	2.0 V					J1			
			14						A			2.0 V	16 mA		0.8 V					K1			
			15							0.8 V	2.0 V		16 mA							J2			
			16							2.0 V	0.8 V			16 mA						K2			
	V_{IC}		17				-12mA												-12mA	Clock 1		-1.5	
			18																	Preset 1			
			19									-12mA								Clear 1			
			20												-12 mA					Clock 2			
			21	-12mA																Preset 2			
			22		-12mA															Clear 2			
			23			-12mA														J1			
			24						-12mA											K1			
			25							-12 mA										J2			
			26								-12 mA									K2			
	I_{IL1}	3009	27	4.5 V		B	0.4 V	5.5 V												J1	-0.7	-1.6	mA
			28	4.5 V	B	4.5 V													0.4 V	K1			
			29						4.5 V		B	0.4 V								J2			
			30						4.5 V	B	4.5 V				0.4 V					K2			
	I_{IL2}		31	0.4 V	B		4.5 V												4.5 V	Clock 1	-1.4	-3.2	
			32	0.4 V		B	4.5 V												4.5 V	Clock 1			
			33						0.4 V	B		4.5 V			4.5 V					Clock 2			
			34						0.4 V		B	4.5 V			4.5 V					Clock 2			
	I_{IL3}		35 A, D	4.5 V		0.4 V	4.5 V												4.5 V	Clear 1	-0.7	-1.6	
			35 B, C	4.5 V		0.4 V	4.5 V												4.5 V	Clear 1	-1.4	-3.2	
			36 A, D	4.5 V	0.4 V		4.5 V												4.5 V	Preset 1	-0.7	-1.6	
			36 B, C	4.5 V	0.4 V		4.5 V												4.5 V	Preset 1	-1.4	-3.2	
			37 A, D						4.5 V		0.4 V	4.5 V			4.5 V					Clear 2	-0.7	-1.6	
			37 B, C						4.5 V		0.4 V	4.5 V			4.5 V					Clear 2	-1.4	-3.2	
			38 A, D						4.5 V	0.4 V		4.5 V			4.5 V					Preset 2	-0.7	-1.6	
			38 B, C						4.5 V	0.4 V		4.5 V			4.5 V					Preset 2	-1.4	-3.2	
	I_{IH1}	3010	39	GND	B	GND	2.4 V													J1	0	40	µA
			40	GND	GND	B													2.4 V	K1			
			41						GND	B	GND	2.4 V								J2			
			42						GND	GND	B				2.4 V					K2			
	I_{IH2}		43	GND	B	GND	5.5 V													J1		100	
			44	GND	GND	B													5.5 V	K1			
			45						GND	B	GND	5.5 V								J2			
			46						GND	GND	B				5.5 V					K2			

See notes at end of device type 04.

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TABLE III. Group A inspection for device type 04 1/ - Continued.

Subgroup	Symbol	MIL-STD-883 method	Case E & F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits Min	Test limits Max	Test limits Unit
				Clock 1	Preset 1	Clear 1	J1	V_{CC}	Clock 2	Preset 2	Clear 2	J2	$\overline{Q}2$	Q2	K2	GND	$\overline{Q}1$	Q1	K1				
1 T_A = 25°C	I_{IH3}	3010	47	GND		E	GND	5.5 V								GND			4.5 V	Clear 1	0	100	µA
			48	GND	E		4.5 V												GND	Preset 1			
			49						GND		E	GND			4.5 V					Clear 2			
			50						GND	E		4.5 V			GND					Preset 2			
	I_{IH4}		51	GND		F	GND												4.5 V	Clear 1		200	
			52	GND	F		4.5 V												GND	Preset 1			
			53						GND		F	GND			4.5 V					Clear 2			
			54						GND	F		4.5 V			GND					Preset 2			
			55	5.5 V	GND		GND												GND	Clock 1			
			56						5.5 V	GND		GND			GND					Clock 2			
	I_{IH5}		57 CKT A, C	2.4 V	GND		GND												GND	Clock 1	-500	-50	
			57 CKT B	2.4 V	GND		GND												GND	Clock 1	-850	-200	
			58 CKT A, C	2.4 V		GND	GND												GND	Clock 1	-500	-50	
			58 CKT B	2.4 V		GND	GND												GND	Clock 1	-850	-200	
			59 CKT A, C						2.4 V	GND		GND			GND					Clock 2	-500	-50	
			59 CKT B						2.4 V	GND		GND			GND					Clock 2	-850	-200	
			60 CKT A, C						2.4 V		GND	GND			GND					Clock 2	-500	-50	
			60 CKT B						2.4 V		GND	GND			GND					Clock 2	-850	-200	
	I_{OS}	3011	61**	2.4 V	GND	4.5 V	2.4 V											GND	2.4 V	$\overline{Q}1$	-30	-57	mA
			62	2.4 V	4.5 V	GND	2.4 V										GND		2.4 V	Q1			
			63**						2.4 V	GND	4.5 V	2.4 V		GND	2.4 V					$\overline{Q}2$			
			64						2.4 V	4.5 V	GND	2.4 V	GND		2.4 V					Q2			
	I_{CC}	3005	65	GND	4.5 V	GND	GND		GND	4.5 V	GND	GND			GND				GND	V_{CC}		60	
	I_{CC}	3005	66	GND	GND	4.5 V	GND		GND	GND	4.5 V	GND			GND				GND	V_{CC}		40	
2	Same tests, terminal conditions and limits as for subgroup 1, except T_A = 125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions and limits as for subgroup 1, except T_A = -55°C and V_{IC} tests are omitted.																						
7 2/ 4/ T_A = 25°C			67	B	A	B	A	4.5 V	B	A	B	A	H 3/	L 3/	B	GND	H 3/	L 3/	B	All output	H or L as shown 3/		
			68	A	A	B	A		A	A	B	A	H	L	B		H	L	B				
			69	B	A	B	A		B	A	B	A	H	L	B		H	L	B				
			70	B	A	B	A		B	A	B	A	H	L	A		H	L	A				
			71	A	A	B	A		A	A	B	A	H	L	A		H	L	A				
			72	B	A	B	A		B	A	B	A	H	L	A		H	L	A				
			73	B	B	A	A		B	B	A	A	L	H	A		L	H	A				
			74	A	B	A	A		A	B	A	A	L	H	A		L	H	A				
			75	B	B	A	A		B	B	A	A	L	H	A		L	H	A				
			76	B	B	A	B		B	B	A	B	L	H	A		L	H	A				
			77	A	B	A	B		A	B	A	B	L	H	A		L	H	A				
			78	B	B	A	B		B	B	A	B	L	H	A		L	H	A				
			79	B	A	A	B		B	A	A	B	L	H	B		L	H	B				
			80	A	A	A	B		A	A	A	B	L	H	B		L	H	B				
			81	B	A	A	B		B	A	A	B	L	H	B		L	H	B				
			82	B	A	B	B		B	A	B	B	H	L	B		H	L	B				
			83	B	A	A	B		B	A	A	B	H	L	B		H	L	B				
			84	A	A	A	B		A	A	A	B	H	L	B		H	L	B				
			85	B	A	A	B		B	A	A	B	H	L	B		H	L	B				
			86	B	A	A	A		B	A	A	A	H	L	B		H	L	B				
			87	A	A	A	A		A	A	A	A	H	L	B		H	L	B				
			88	B	A	A	A		B	A	A	A	L	H	B		L	H	B				

See notes at end of device type 04.

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TABLE III. Group A inspection for device type 04 1/ - Continued.

Subgroup	Symbol	MIL-STD-883 method	Case E & F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits		
				Clock 1	Preset 1	Clear 1	J1	V_{CC}	Clock 2	Preset 2	Clear 2	J2	Q2	$\overline{Q2}$	K2	GND	Q1	$\overline{Q1}$	K1		Min	Max	Unit
7 2/ 4/ T_A = 25 C			89	B	A	A	B	4.5 V	B	A	A	B	L 3/	H 3/	A	GND	L 3/	H 3/	A	All output	H or L as shown 3/		
			90	A	A	A	B		A	A	A	B	L	H	A		L	H	A				
			91	B	A	A	B		B	A	A	B	H	L	A		H	L	A				
			92	B	A	A	A		B	A	A	A	H	L	A		H	L	A				
			93	A	A	A	A		A	A	A	A	H	L	A		H	L	A				
			94	B	A	A	A		B	A	A	A	L	H	A		L	H	B				
			95	A	A	A	A		A	A	A	A	L	H	A		L	H	A				
			96	B	A	A	A		B	A	A	A	H	L	A		H	L	A				
			97	A	B	B	B		A	B	B	B	H	H	B		H	H	B				
			98	A	B	A	B		A	B	A	A	L	H	A		L	H	A				
			99	A	A	A	A		A	A	A	A	L	H	A		L	H	A				
			100	A	A	A	A		A	A	A	A	L	H	B		L	H	B				
			101	A	A	A	B		A	A	A	B	L	H	B		L	H	B				
			102	B	A	A	B		B	A	A	B	H	L	B		H	L	B				
			103	A	A	A	A		A	A	A	A	H	L	A		H	L	A				
			104	A	A	A	B		A	A	A	B	H	L	A		H	L	A				
			105	A	A	A	B		A	A	A	B	H	L	B		H	L	B				
			106	B	A	A	B		B	A	A	B	L	H	B		L	H	B				
8 2/ 4	Same tests, terminal conditions and limits as for subgroup 7, except T_A = 125 and -55 C.																						
9 T_A = 25 C	F_{MAX}	(Fig 10)	107	IN	5.0 V	5.0 V	2.4 V	5.0 V								GND	OUT	OUT	2.4 V	Q1	5		MHz
			108	IN	5.0 V	5.0 V	2.4 V												2.4 V	$\overline{Q1}$			
			109						IN	5.0 V	5.0 V	2.4 V		OUT	2.4 V					Q2			
			110						IN	5.0 V	5.0 V	2.4 V	OUT		2.4 V					$\overline{Q2}$			
	t_{PLH1}	3003 (Fig 9)	111	2.4 V	J	IN	2.4 V										OUT		2.4 V	Clear 1 to $\overline{Q1}$	5	25	ns
			112	2.4 V	IN	J	2.4 V											OUT	2.4 V	Preset 1 to Q1			
			113						2.4 V	J	IN	2.4 V	OUT		2.4 V					Clear 2 to $\overline{Q2}$			
			114						2.4 V	IN	J	2.4 V		OUT	2.4 V					Preset 2 to Q2			
	t_{PHL1}		115	2.4 V	J	IN	2.4 V											OUT	2.4 V	Clear 1 to Q1		40	
			116	2.4 V	IN	J	2.4 V										OUT		2.4 V	Preset 1 to $\overline{Q1}$			
			117						2.4 V	J	IN	2.4 V		OUT	2.4 V					Clear 2 to Q2			
			118						2.4 V	IN	J	2.4 V	OUT		2.4 V					Preset 2 to $\overline{Q2}$			
	t_{PLH2}	3003 (Fig 10)	119	IN	5.0 V	5.0 V	2.4 V											OUT	2.4 V	Clock 1 to Q1	5	30	ns
			120	IN	5.0 V	5.0 V	2.4 V										OUT		2.4 V	Clock 1 to $\overline{Q1}$			
			121						IN	5.0 V	5.0 V	2.4 V		OUT	2.4 V					Clock 2 to Q2			
			122						IN	5.0 V	5.0 V	2.4 V	OUT		2.4 V					Clock 2 to $\overline{Q2}$			
	t_{PHL2}		123	IN	5.0 V	5.0 V	2.4 V											OUT	2.4 V	Clock 1 to Q1		40	

See notes at end of device type 04.

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TABLE III. Group A inspection for device type 04 1/ - Continued.

Subgroup	Symbol	MIL-STD-883 method	Case F & F	1 Clock 1	2 Preset 1	3 Clear 1	4 J1	5 V_{CC}	6 Clock 2	7 Preset 2	8 Clear 2	9 J2	10 $\overline{Q2}$	11 Q2	12 K2	13 GND	14 $\overline{Q1}$	15 Q1	16 K1	Meas. terminal	Test limits Min	Max	Unit
9	t_{PHL2}	3003 (Fig 10)	124	IN	5.0 V	5.0 V	2.4 V	5.0 V								GND	OUT		2.4 V	Clock 1 to $\overline{Q1}$	5	40	ns
T_A 25°C			125						IN	5.0 V	5.0 V	2.4 V		OUT	2.4 V					Clock 2 to Q2			
			126						IN	5.0 V	5.0 V	2.4 V	OUT		2.4 V					Clock 2 to $\overline{Q2}$			
10	F_{MAX}	(Fig 10)	127	IN	5.0 V	5.0 V	2.4 V												2.4 V	$\overline{Q1}$	5		MHz
T_A 125°C			128	IN	5.0 V	5.0 V	2.4 V										OUT		2.4 V	$\overline{Q1}$			
			129						IN	5.0 V	5.0 V	2.4 V		OUT	2.4 V					Q2			
			130						IN	5.0 V	5.0 V	2.4 V	OUT		2.4 V					$\overline{Q2}$			
	t_{PLH1}	3003 (Fig 9)	131	2.4 V	J	IN	2.4 V										OUT		2.4 V	Clear 1 to $\overline{Q1}$	5	39	ns
			132	2.4 V	IN	J	2.4 V											OUT	2.4 V	Preset 1 to Q1			
			133						2.4 V	J	IN	2.4 V	OUT		2.4 V					Clear 2 to Q2			
			134						2.4 V	IN	J	2.4 V		OUT	2.4 V					Preset 2 to Q2			
	t_{PHL1}		135	2.4 V	J	IN	2.4 V											OUT	2.4 V	Clear 1 to Q1		50	
			136	2.4 V	IN	J	2.4 V										OUT		2.4 V	Preset 1 to $\overline{Q1}$			
			137						2.4 V	J	IN	2.4 V		OUT	2.4 V					Clear 2 to Q2			
			138						2.4 V	IN	J	2.4 V	OUT		2.4 V					Preset 2 to $\overline{Q2}$			
	t_{PLH2}	3003 (Fig 10)	139	IN	5.0 V	5.0 V	2.4 V											OUT	2.4 V	Clock 1 to Q1	5	39	ns
			140	IN	5.0 V	5.0 V	2.4 V										OUT		2.4 V	Clock 1 to $\overline{Q1}$			
			141						IN	5.0 V	5.0 V	2.4 V		OUT	2.4 V					Clock 2 to Q2			
			142						IN	5.0 V	5.0 V	2.4 V	OUT		2.4 V					Clock 2 to $\overline{Q2}$			
	t_{PHL2}		143	IN	5.0 V	5.0 V	2.4 V											OUT	2.4 V	Clock 1 to Q1		50	
			144	IN	5.0 V	5.0 V	2.4 V										OUT		2.4 V	Clock 1 to $\overline{Q1}$			
			145						IN	5.0 V	5.0 V	2.4 V		OUT	2.4 V					Clock 2 to Q2			
			146						IN	5.0 V	5.0 V	2.4 V	OUT		2.4 V					Clock 2 to $\overline{Q2}$			
11	Same tests, terminal conditions and limits as subgroup 10, except T_A = -55°C.																						

NOTES:
- A Normal clock pulse.
- B Momentary GND, then 4.5 V.
- C Momentary GND, then open.
- E Momentary ground, then 2.4 V.
- F Momentary ground, then 5.5 V.
- ** Test time limit ≥ 100 ms.
- J Input pulse t_p = 100 ns, PRR = 1 MHz, V_{OL} = 0 V, V_{OH} = 4.5 V.

1 Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open).
2 Input voltages shown are: A = 2.0 volts minimum and B = 0.8 volts maximum.
3 Output voltages shall be either: (a) H = 2.4 V, minimum and L = 0.4 V, maximum when using a high speed checker double comparator, or (b) H ≥ 1.5 V and L < 1.5 V when using a high speed checker single comparator.
4 Tests shall be performed in sequence.

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TABLE III. Group A inspection for device type 05 1/

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D Case C	1 3	2 2	3 1	4 14	5 13	6 12	7 11	8 10	9 9	10 8	11 7	12 6	13 5	14 4	Meas. terminal	Test limits Min	Max	Unit
			Test No.	Clock 1	D1	Clear 1	V_{CC}	Clear 2	D2	Clock 2	Preset 2	Q_2	$\overline{Q}_2$	GND	$\overline{Q}_1$	Q_1	Preset 1				
1 T_A = 25°C	V_{OH}	3006	1	A	2.0 V		4.5 V							GND		-.4mA		Q_1	2.4		V
			2	A	0.8 V										-.4 mA			$\overline{Q}_1$			
			3	GND		0.8 V									-.4 mA		GND	$\overline{Q}_1$			
			4			GND											0.8 V	Q_1			
			5						2.0 V	A		-.4mA				-.4mA		Q_2			
			6						0.8 V	A			-.4 mA					$\overline{Q}_2$			
			7					0.8 V		GND	GND		-.4 mA					Q_2			
			8					GND			0.8 V	-.4mA						$\overline{Q}_2$			
	V_{OL}	3007	9	A	2.0 V										16 mA	16 mA		Q_1		0.4	
			10	A	0.8 V											16 mA	2.0 V	$\overline{Q}_1$			
			11			0.8 V			2.0 V	A			16 mA		16 mA		0.8 V	Q_1			
			12			2.0 V			0.8 V	A		16mA						$\overline{Q}_1$			
			13					0.8 V			2.0 V	16mA						Q_2			
			14					2.0 V			0.8 V							$\overline{Q}_2$			
			15										16 mA					Q_2			
			16										16 mA					$\overline{Q}_2$			
	V_{IC}		17		-12mA													D1		-1.5	
			18	-12 mA														Clock 1			
			19			-12mA												Clear 1			
			20						-12mA								-12mA	Preset 1			
			21							-12mA								D2			
			22															Clock 2			
			23					-12mA										Clear 2			
			24								-12 mA							Preset 2			
	I_{IL1}	3009	25	4.5 V	0.4 V	4.5 V	5.5 V														
			26					4.5 V	0.4 V	4.5 V	GND						GND	D1	-0.7	-1.6	mA
			27	GND	GND	GND												D2	-0.7	-1.6	
			28					GND	GND	GND	0.4 V						0.4 V	Preset 1	-1.4	-3.2	
	I_{IL2}		29	0.4 V	GND	4.5 V												Preset 2	-1.4	-3.2	
			30	4.5 V	4.5 V	0.4 V											GND	Clock 1	-1.4	-3.2	
			31					4.5 V	GND	0.4 V	GND						GND	Clear 1	-2.1	-4.8	
			32					0.4 V	4.5 V	4.5 V	GND							Clock 2	-1.4	-3.2	
	I_{IH1}	3010	33	4.5 V	2.4 V	GND												Clear 2	-2.1	-4.8	
	I_{IH1}		34					GND	2.4 V	4.5 V	4.5 V						4.5 V	D1		40	μA
	I_{IH2}		35	4.5 V	5.5 V	GND												D2		40	
	I_{IH2}		36					GND	5.5 V	4.5 V	4.5 V						4.5 V	D1		100	
	I_{IH3}		37	2.4 V	4.5 V	GND												D2		100	
			38	B	4.5 V	4.5 V											4.5 V	Clock 1			
			39					GND	4.5 V	2.4 V	4.5 V						2.4 V	Preset 1		80	
			40					4.5 V	4.5 V	B	2.4 V							Clock 2			
	I_{IH4}		41	5.5 V	4.5 V	GND												Preset 2			
			42	B	4.5 V	4.5 V											4.5 V	Clock 1			
			43					GND	4.5 V	5.5 V	4.5 V						5.5 V	Preset 1		200	
			44					4.5 V	4.5 V	B	5.5 V							Clock 2			
	I_{IH5}		45	B	GND	2.4 V												Preset 2			
	I_{IH5}		46					2.4 V	GND	B	4.5 V						4.5 V	Clear 1		120	
	I_{IH6}		47	B	GND	5.5 V												Clear 2		120	
	I_{IH6}		48					5.5 V	GND	B	4.5 V						4.5 V	Clear 1		300	
																		Clear 2		300	

See notes at end of device type 05.

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TABLE III Group A inspection for device type 05 [1] - Continued.

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D / Case C / Test No.	1 / 3 / Clock 1	2 / 2 / D1	3 / 1 / Clear 1	4 / 14 / V_{CC}	5 / 13 / Clear 2	6 / 12 / D2	7 / 11 / Clock 2	8 / 10 / Preset 2	9 / 9 / Q_2	10 / 8 / $\overline{Q}_2$	11 / 7 / GND	12 / 6 / $\overline{Q}_1$	13 / 5 / Q_1	14 / 4 / Preset 1	Meas. terminal	Test limits Min	Max	Unit
1 T_A = 25°C	I_{OS}	3011	49			GND	5.5 V				GND	GND	GND	GND	GND	GND	GND	Q_1	-20	-57	mA
			50															$\overline{Q}_1$			
			51															Q_2			
			52					GND					GND					$\overline{Q}_2$			
	I_{CC}	3005	53	GND	GND	GND			GND	GND	GND			GND			GND	V_{CC}		30	
	I_{CC}	3005	54	GND	GND			GND	GND	GND								V_{CC}		30	
2	Same tests, terminal conditions and limits as subgroup 1, except T_A = 125° C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as subgroup 1, except T_A = -55° C and V_{IC} tests are omitted.																				
7 2/ 4/ T_A = 25°C			55	B	B	B	4.5 V	B	B	B	B	H 3/	H 3/	GND	H 3/	H	B	All output	H or L as shown 3/		
			56	B	B	B		B	B	B	A	L	H		H	L	A				
			57	B	B	A		A	B	B	A	L	H		L	H	B				
			58	B	B	A		A	B	B	B	H	L		L	H	B				
			59	A	B	A		A	B	A	B	H	L		H	H	B				
			60	A	B	B		B	B	A	B	H	H		H	H	A				
			61	A	A	B		B	A	A	A	L	H		H	L	A				
			62	A	A	B		A	A	A	B	H	L		L	H	A				
			63	A	A	A		A	A	A	A	H	L		L	H	B				
			64	A	A	A		A	A	A	A	H	L		H	H	B				
			65	A	A	A		A	A	B	A	H	L		H	H	A				
			66	B	A	A		A	B	B	A	L	H		L	H	A				
			67	B	B	A		A	B	B	B	H	L		L	H	A				
			68	A	B	A		A	B	A	B	L	H		L	H	B				
			69	A	B	A		A	B	A	B	H	L		H	H	B				
			70	A	B	B		B	B	A	A	L	H		H	L	A				
			71	A	B	B		B	B	A	A	H	L		L	L	A				
			72	A	B	B		B	B	A	A	L	H		L	L	A				
			73	A	B	A		A	B	A	A	H	L		H	L	A				
			74	B	A	A		A	A	A	B	L	H		H	L	B				
			75	A	A	A		A	A	A	A	H	L		L	H	B				
			76	A	A	A		A	A	A	A	L	H		L	H	A				
			77	A	A	A		A	A	A	A	H	L		H	H	A				
			78	A	A	B		A	A	A	B	L	H		H	L	B				
			79	A	A	A		A	A	B	A	H	L		L	L	B				
			80	A	B	A		A	B	A	A	L	H		L	L	A				
			81	A	B	A		A	B	A	B	H	L		H	L	A				
8 2/ 4/	Same tests, terminal conditions and limits as for subgroup 7, except T_A = 125 and -55° C.																				
9 T_A = 25°C	F_{MAX}	(Fig 12)	82	IN	E	5.0 V	5.0 V		E	IN	5.0 V	OUT		GND	OUT	OUT	5.0 V	Q_1	5		MHz
			83	IN	E	5.0 V		5.0 V	E	IN	5.0 V		OUT				5.0 V	$\overline{Q}_1$			
			84															Q_2			
			85															$\overline{Q}_2$			

See notes at end of device type 05.

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TABLE III. Group A inspection for device type 05 1/ - Continued.

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D / Case C / Test No.	1 / 3 / Clock 1	2 / 2 / D1	3 / 1 / Clear 1	4 / 14 / V_{CC}	5 / 13 / Clear 2	6 / 12 / D2	7 / 11 / Clock 2	8 / 10 / Preset 2	9 / 9 / Q2	10 / 8 / $\overline{Q2}$	11 / 7 / GND	12 / 6 / $\overline{Q1}$	13 / 5 / Q1	14 / 4 / Preset 1	Meas. terminal	Test limits Min	Max	Unit
9 T_A = 25°C	t_{PLH1}	3003 (Fig 11)	86			IN	5.0 V							GND	OUT		J	Clear 1 to $\overline{Q1}$	5	25	ns
			87			J										OUT	IN	Preset 1 to Q1			
			88					IN			J		OUT					Clear 2 to $\overline{Q2}$			
			89					J			IN	OUT						Preset 2 to Q2			
	t_{PHL1}		90			IN										OUT	J	Clear 1 to Q1		40	
			91			J									OUT		IN	Preset 1 to $\overline{Q1}$			
			92					IN			J	OUT						Clear 2 to Q2			
			93					J			IN		OUT					Preset 2 to $\overline{Q2}$			
	t_{PLH2}	3003 (Fig 12) (Fig 13)	94	IN	IN	5.0 V										OUT	5.0 V	Clock 1 to Q1	5	30	ns
			95	IN	IN	5.0 V									OUT		5.0 V	Clock 1 to $\overline{Q1}$			
		(Fig 12)	96					5.0 V	IN	IN	5.0 V	OUT						Clock 2 to Q2			
		(Fig 13)	97					5.0 V	IN	IN	5.0 V		OUT					Clock 2 to $\overline{Q2}$			
	t_{PHL2}	(Fig 13)	98	IN	IN	5.0 V										OUT	5.0 V	Clock 1 to Q1		40	
		(Fig 12)	99	IN	IN	5.0 V									OUT		5.0 V	Clock 1 to $\overline{Q1}$			
		(Fig 13)	100					5.0 V	IN	IN	5.0 V	OUT						Clock 2 to Q2			
		(Fig 12)	101					5.0 V	IN	IN	5.0 V		OUT					Clock 2 to $\overline{Q2}$			
10 T_A = 125°C	F_{MAX}	(Fig 12)	102	IN	E	5.0 V	5.0 V									OUT	5.0 V	Q1	5		MHz
			103	IN	E	5.0 V									OUT		5.0 V	$\overline{Q1}$			
			104					5.0 V	E	IN	5.0 V	OUT						Q2			
			105					5.0 V	E	IN	5.0 V		OUT					$\overline{Q2}$			
	t_{PLH1}	3003 (Fig 11)	106			IN									OUT		J	Clear 1 to $\overline{Q1}$	5	35	ns
			107			J										OUT	IN	Preset 1 to Q1			
			108					IN			J		OUT					Clear 2 to $\overline{Q2}$			
			109					J			IN	OUT						Preset 2 to Q2			
	t_{PHL1}		110			IN										OUT	J	Clear 1 to Q1		50	
			111			IN									OUT		IN	Preset 1 to $\overline{Q1}$			
			112					IN			J	OUT						Clear 2 to Q2			
			113					J			IN		OUT					Preset 2 to $\overline{Q2}$			

See notes at end of device type 05.

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TABLE III. Group A inspection for device type 05 1/ - Continued.

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D Case C Test No.	1 3 Clock 1	2 2 D1	3 1 Clear 1	4 14 V_{CC}	5 13 Clear 2	6 12 D2	7 11 Clock 2	8 10 Preset 2	9 9 Q2	10 8 $\overline{Q}$2	11 7 GND	12 6 $\overline{Q}$1	13 5 Q1	14 4 Preset 1	Meas. terminal	Test limits Min	Max	Unit
10 T_A = 125°C	t_{PLH2}	3003 5/ (Fig 12)	114	IN	IN	5.0 V	5.0 V							GND		OUT	5.0 V	Clock 1 to Q1	5	39	ns
		(Fig 13)	115	IN	IN	5.0 V									OUT		B	Clock 1 to $\overline{Q}$1			
		(Fig 12)	116					B	IN	IN	5.0 V	OUT						Clock 2 to Q2			
		(Fig 13)	117					5.0 V	IN	IN	5.0 V		OUT					Clock 2 to $\overline{Q}$2			
	t_{PHL2}	(Fig 13)	118	IN	IN	5.0 V										OUT	B	Clock 1 to Q1		40	
		(Fig 12)	119	IN	IN	5.0 V									OUT		5.0 V	Clock 1 to $\overline{Q}$1			
		(Fig 13)	120					5.0 V	IN	IN	5.0 V	OUT						Clock 2 to Q2			
		(Fig 12)	121					B	IN	IN	5.0 V		OUT					Clock 2 to $\overline{Q}$2			
11	Same tests, terminal conditions and limits as subgroup 10, except T_A = -55°C.																				

NOTES:
A = Normal clock pulse.
B = Momentary GND, then 4.5 V.
E = Input D connected to $\overline{Q}$.
J = Input pulse, tp $\geq$ 100 ns, PRR = 1 MHz, V_{OL} = 0 V, V_{OH} = 4.5 V.

1/ Terminal conditions (pins not designated may be H $\geq$ 2.0 V, or L $\leq$ 0.8 V, or open).
2/ Input voltages shown are: A = 2.0 volts minimum and B = 0.8 volts maximum.
3/ Output voltages shall be either: (a) H = 2.4 V, minimum and L = 0.4 V, maximum when using a high speed checker double comparator; or (b) H $\geq$ 1.5 V and L < 1.5 V when using a high speed speed checker single comparator.
4/ Tests shall be performed in sequence.
5/ Tests shall be performed for both D input pulses (A and B).

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TABLE III. Group A inspection for device type 06 1/

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas. terminal	Test limits		
			Case C	10	12	13	14	2	1	3	4	5	6	7	8	9	11				
			Test No.	K1	Clock	Preset	VCC	Clear	NC	J1	J2	J*	$\overline{Q}$	GND	Q	K*	K2		Min	Max	Unit
1 T_A = 25° C	V_{OH}	3006	1	0.8 v	A		4.5 v			2.0 v	2.0 v	0.8 v		GND	-.4 mA	2.0 v	0.8 v	Q	2.4		V
			2	2.0 v	A					0.8 v	0.8 v	2.0 v	-.4 mA			0.8 v	2.0 v	$\overline{Q}$			
			3		GND	2.0 v		0.8 v				GND	-.4 mA			GND		Q			
			4		GND	0.8 v		2.0 v				GND			-.4 mA	GND		$\overline{Q}$			
	V_{OL}	3007	5	2.0 v	A					0.8 v	0.8 v	2.0 v			16 mA	0.8 v	2.0 v	Q		0.4	
			6	0.8 v	A					2.0 v	2.0 v	0.8 v	16 mA			2.0 v	0.8 v	$\overline{Q}$			
			7		GND	0.8 v		2.0 v				GND	16 mA					Q			
			8		GND	2.0 v		0.8 v				GND			16 mA			$\overline{Q}$			
	V_{IC}		9							-12mA								J1		-1.5	
			10								-12 mA							J2			
			11									-12 mA						J*			
			12	-12 mA														K1			
			13														-12 mA	K2			
			14													-12 mA		K*			
			15		-12 mA													Clock			
			16			-12 mA												Preset			
			17					-12 mA										Clear			
	I_{IL1}	3009	18		GND		5.5 v	B		0.4 v	4.5 v	0.4 v						J1	-0.7	-1.6	mA
			19		GND			B		4.5 v	0.4 v	0.4 v						J2			
			20									0.4 v						J*			
			21	0.4 v	GND	B										0.4 v	4.5 v	K1			
			22	4.5 v	GND	B										0.4 v	0.4 v	K2			
			23													0.4 v		K*			
			24		0.4 v													Clock			
			25	4.5 v	GND	0.4 v										0.4 v	4.5 v	Preset			
			26		GND			0.4 v		4.5 v	4.5 v	0.4 v						Clear			
	I_{IH1}	3010	27					GND		2.4 v	GND	4.5 v						J1	0	40	μA
			28					GND		GND	2.4 v	4.5 v						J2			
			29									2.4 v						J*			
			30	2.4 v		GND										4.5 v	GND	K1			
			31	GND		GND										4.5 v	2.4 v	K2			
			32													2.4 v		K*			
			33		2.4 v													Clock			
	I_{IH2}		34					GND		5.5 v	GND	4.5 v						J1		100	
			35					GND		GND	5.5 v	4.5 v						J2			
			36									5.5 v						J*			
			37	5.5 v		GND										4.5 v	GND	K1			
			38	GND		GND										4.5 v	5.5 v	K2			
			39													5.5 v		K*			
			40		5.5 v													Clock			
	I_{IH3}		41	GND	A	2.4 v				4.5 v	4.5 v	GND				4.5 v	GND	Preset		80	
			42	4.5 v	A			2.4 v		GND	GND	4.5 v				GND	4.5 v	Clear		80	
	I_{IH4}		43	GND	A	5.5 v				4.5 v	4.5 v	GND				4.5 v	GND	Preset		200	
			44	4.5 V	A			5.5 v		GND	GND	4.5 v				GND	4.5 v	Clear		200	
	I_{OS}	3011	45		GND	GND						GND			GND	GND		Q	-20	-57	mA
			46		GND			GND				GND	GND			GND		$\overline{Q}$	-20	-57	
	I_{CC}	3005	47					GND										VCC		30	
			48			GND												VCC		30	
2	Same tests, terminal conditions and limits as for subgroup 1, except T_A = 125° C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_A = -55° C and V_{IC} tests are omitted.																				

See notes at end of device type 06.

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TABLE III. Group A inspection for device type 06 1/ - Continued.

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D / Case C / Test No.	1 / 10 / K1	2 / 12 / Clock	3 / 13 / Preset	4 / 14 / V_{CC}	5 / 2 / Clear	6 / 1 / NC	7 / 3 / J1	8 / 4 / J2	9 / 5 / $\overline{J}$	10 / 6 / $\overline{Q}$	11 / 7 / GND	12 / 8 / Q	13 / 9 / $\overline{K}$	14 / 11 / K2	Meas. terminal	Test limits: Min	Max	Unit
7 2/ 4/ T_A = 25°C			49	B	B	A	4.5 v	B	B	B	B	A	H 3/	GND	L 3/	A	B	All output	H or L as shown 3/		
			50	B	B	B		A	B	B	B	A	L		H	A	B				
			51	B	B	A		A	B	B	B	A	L		H	B	A				
			52	B	A	A		A	B	B	B	A	L		H	B	A				
			53	B	B	A		A	B	B	B	A	L		H	B	A				
			54	A	B	A		A	B	B	B	A	L		H	B	B				
			55	A	A	A		A	B	B	B	A	L		H	B	B				
			56	A	B	A		A	B	B	B	A	L		H	B	B				
			57	A	B	A		A	B	B	B	A	L		H	A	A				
			58	A	A	A		A	B	B	B	A	L		H	A	A				
			59	A	B	A		A	B	B	B	A	L		H	A	A				
			60	A	B	A		B	B	B	B	A	H		L	A	A				
			61	B	B	A		A	B	A	B	B	H		L	A	B				
			62	B	A	A		A	B	A	B	B	H		L	A	B				
			63	B	B	A		A	B	A	B	B	H		L	A	B				
			64	B	B	A		A	B	B	A	B	H		L	A	B				
			65	B	A	A		A	B	B	A	B	H		L	A	B				
			66	B	B	A		A	B	B	A	B	H		L	A	B				
			67	B	B	A		A	B	A	A	A	H		L	A	B				
			68	B	A	A		A	B	A	A	A	H		L	A	B				
			69	B	B	A		A	B	A	A	A	H		L	A	B				
			70	A	B	A		A	B	A	A	B	H		L	B	A				
			71	A	A	A		A	B	A	A	B	L		H	B	A				
			72	A	B	A		A	B	A	A	B	L		H	B	A				
			73	A	A	A		A	B	A	A	B	H		L	B	A				
			74	A	B	A		A	B	A	A	B	H		L	B	A				
			75	A	B	A		B	B	A	A	B	H		L	B	A				
			76	A	A	A		B	B	A	A	B	H		L	B	A				
			77	A	B	A		B	B	A	A	B	H		L	B	A				
			78	A	B	B		B	B	A	A	B	L		L	B	A				
			79	A	A	B		B	B	A	A	B	L		L	B	A				
			80	A	B	B		B	B	A	A	B	L		L	B	A				
			81	B	B	A		B	B	B	B	A	H		L	A	B				
			82	B	B	A		A	B	B	B	A	H		L	A	B				
			83	B	A	A		A	B	B	B	A	H		L	A	B				
			84	B	A	B		A	B	B	B	A	L		L	A	B				
			85	B	B	B		A	B	B	B	A	L		H	A	B				
			86	B	B	A		A	B	B	B	A	L		H	A	B				
			87	B	A	A		A	B	B	B	A	L		H	A	B				
			88	B	A	A		B	B	B	B	A	L		L	A	B				
			89	B	A	A		A	B	B	B	A	L		H	A	B				
			90	A	B	A		A	B	A	A	B	L		H	B	A				
			91	A	B	B		A	B	A	A	B	L		H	B	A				
			92	A	A	B		A	B	A	A	B	L		H	B	A				
8 2/ 4/	Same tests, terminal conditions and limits as for subgroup 7, except T_A = 125 and -55° C.																				

See notes at end of device type 06.

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TABLE III. Group A inspection for device type 06 1/ - Continued.

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D / Case C / Test No.	1 / 10 / K1	2 / 12 / Clock	3 / 13 / Preset	4 / 14 / VCC	5 / 2 / Clear	6 / 1 / NC	7 / 3 / J1	8 / 4 / J2	9 / 5 / J*	10 / 6 / $\overline{Q}$	11 / 7 / GND	12 / 8 / Q	13 / 9 / K*	14 / 11 / K2	Meas. terminal	Test limits Min	Max	Unit
9 T_A = 25°C	F_{MAX}	(Fig. 15)	93	2.4 v	IN	5.0 v	5.0 v	5.0 v		2.4 v	2.4 v	GND		GND	OUT	GND	2.4 v	Q	10		MHz
	F_{MAX}	(Fig. 15)	94	2.4 v	IN	5.0 v		5.0 v		2.4 v	2.4 v	GND	OUT			GND	2.4 v	$\overline{Q}$	10		MHz
	t_{PLH}	3003	95	5.0 v	0.8 v	IN		IN		5.0 v	5.0 v	GND	OUT			GND	5.0 v	Clear to $\overline{Q}$	5	50	ns
	t_{PLH}	(Fig. 14)	96												OUT			Preset to Q			
	t_{PHL}		97												OUT			Clear to Q			
	t_{PHL}		98										OUT					Preset to $\overline{Q}$			
	t_{PLH}	3003	99	2.4 v	IN	5.0 v		5.0 v		2.4 v	2.4 v	GND			OUT	GND	2.4 v	Clock to Q	5	50	ns
	t_{PLH}	(Fig. 15)	100										OUT					Clock to $\overline{Q}$			
	t_{PHL}		101												OUT			Clock to Q			
	t_{PHL}		102										OUT					Clock to $\overline{Q}$			
10 T_A = 125°C	F_{MAX}	(Fig. 15)	103	2.4 v	IN	5.0 v		5.0 v		2.4 v	2.4 v	GND			OUT	GND	2.4 v	Q	7.5		MHz
	F_{MAX}	(Fig. 15)	104	2.4 v	IN	5.0 v		5.0 v		2.4 v	2.4 v	GND	OUT			GND	2.4 v	$\overline{Q}$	7.5		MHz
	t_{PLH}	3003	105	5.0 v	0.8 v	IN		IN		5.0 v	5.0 v	GND	OUT			GND	5.0 v	Clear to $\overline{Q}$	5	62	ns
	t_{PLH}	(Fig. 14)	106												OUT			Preset to Q			
	t_{PHL}		107												OUT			Clear to Q			
	t_{PHL}		108										OUT					Preset to $\overline{Q}$			
	t_{PLH}	3003	109	2.4 v	IN	5.0 v		5.0 v		2.4 v	2.4 v	GND			OUT	GND	2.4 v	Clock to Q	5	62	ns
	t_{PLH}	(Fig. 15)	110										OUT					Clock to $\overline{Q}$			
	t_{PHL}		111												OUT			Clock to Q			
	t_{PHL}		112										OUT					Clock to $\overline{Q}$			
11	Same tests, terminal conditions and limits as subgroup 10, except T_A = -55°C.																				

NOTES:
A = Normal clock pulse.
B = Momentary GND, then 4.5 v.

1/ Terminal conditions (pins not designated may be H $\geq$ 2.0 v, or L $\leq$ 0.8 v, or open).
2/ Input voltages shown are: A = 2.0 volts minimum and B = 0.8 volts maximum.
3/ Output voltages shall be either: (a) H = 2.4 v, minimum and L = 0.4 v, maximum when using a high speed checker-double comparator; or (b) H $\geq$ 1.5 v and L < 1.5 v when using a high speed checker single comparator.
4/ Tests shall be performed in sequence.

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TABLE III. Group A inspection for device type 07 1/

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D / Case C / Test No.	1 / 3 / Clock 1	2 / 2 / D1	3 / 1 / Clear 1	4 / 14 / VCC	5 / 13 / Clear 2	6 / 12 / D2	7 / 11 / Clock 2	8 / 10 / Preset 2	9 / 9 / Q_2	10 / 8 / $\overline{Q}_2$	11 / 7 / GND	12 / 6 / $\overline{Q}_1$	13 / 5 / Q_1	14 / 4 / Preset 1	Meas. terminal	Test limits Min	Test limits Max	Unit
1 T_A = 25°C	V_{OH}	3006	1	A	2.0 v		4.5 v							GND		-.4 mA		$\overline{Q}_1$	2.4		V
			2	A	0.8 v										-.4 mA			Q_1			
			3			0.8 v									-.4 mA		2.0 v	$\overline{Q}_1$			
			4			2.0 v										-.4 mA	0.8 v	Q_1			
			5						2.0 v	A		-.4 mA						$\overline{Q}_2$			
			6						0.8 v	A			-.4 mA					Q_2			
			7					0.8 v			2.0 v		-.4 mA					$\overline{Q}_2$			
			8					2.0 v			0.8 v	-.4 mA						Q_2			
	V_{OL}	3007	9	A	2.0 v										16 mA			$\overline{Q}_1$		0.4	
			10	A	0.8 v											16 mA		Q_1			
			11			0.8 v										16 mA	2.0 v	$\overline{Q}_1$			
			12			2.0 v									16 mA		0.8 v	Q_1			
			13						2.0 v	A			16 mA					$\overline{Q}_2$			
			14						0.8 v	A		16 mA						Q_2			
			15					0.8 v			2.0 v	16 mA						$\overline{Q}_2$			
			16					2.0 v			0.8 v		16 mA					Q_2			
	V_{IC}		17		-12 mA													D1		-1.5	
			18	-12 mA														Clock 1			
			19			-12 mA												Clear 1			
			20														-12 mA	Preset 1			
			21						-12 mA									D2			
			22							-12 mA								Clock 2			
			23					-12 mA										Clear 2			
			24								-12 mA							Preset 2			
	I_{IL1}	3009	25	4.5 v	0.4 v	4.5 v	5.5 v										0.4 v	D1	-0.5	-1.6	mA
			26	0.4 v	0.4 v	4.5 v											0.4 v	Preset 1			
			27					4.5 v	0.4 v	4.5 v	0.4 v							D2			
			28					4.5 v	0.4 v	0.4 v	0.4 v							Preset 2			
	I_{IL2}		29	0.4 v	0.4 v	4.5 v											GND	Clock 1	-1.0	-3.2	
			30	0.8 v	4.5 v	0.4 v											4.5 v	Clear 1			
			31					4.5 v	0.4 v	0.4 v	GND							Clock 2			
			32					0.4 v	4.5 v	0.8 v	4.5 v							Clear 2			
	I_{IH1}	3010	33	4.5 v	2.4 v	GND											GND	D1	0	40	μA
			34					GND	2.4 v	4.5 v	GND							D2		40	
	I_{IH2}		35	4.5 v	5.5 v	GND											GND	D1		100	
			36					GND	5.5 v	4.5 v	GND							D2		100	
	I_{IH3}		37	2.4 v		B											GND	Clock 1		80	
			38	B	4.5 v	4.5 v											2.4 v	Preset 1			
			39					B		2.4 v	GND							Clock 2			
			40					4.5 v	4.5 v	B	2.4 v							Preset 2			
	I_{IH4}		41	5.5 v		B											GND	Clock 1		200	
			42	B	4.5 v	4.5 v											5.5 v	Preset 1			
			43					B		5.5 v	GND							Clock 2			
			44					4.5 v	4.5 v	B	5.5 v							Preset 2			
	I_{IH5}		45	GND	GND	2.4 v												Clear 1		120	
			46					2.4 v	GND	GND								Clear 2		120	
	I_{IH6}		47	GND	GND	5.5 v												Clear 1		300	
			48					5.5 v	GND	GND								Clear 2		300	

See notes at end of device type 07.

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TABLE III. Group A inspection for device type 07 1/ - Continued.

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D / Case C / Test No.	1 / 3 / Clock 1	2 / 2 / D1	3 / 1 / Clear 1	4 / 14 / VCC	5 / 13 / Clear 2	6 / 12 / D2	7 / 11 / Clock 2	8 / 10 / Preset 2	9 / 9 / Q_2	10 / 8 / $\overline{Q}_2$	11 / 7 / GND	12 / 6 / $\overline{Q}_1$	13 / 5 / Q_1	14 / 4 / Preset 1	Meas. terminal	Test limits: Min	Max	Unit
1 T_A = 25°C	I_{OS}	3011	49 50 51 52			GND	4.5 v	GND			GND	GND	GND	GND	GND	GND	GND	Q_1 $\overline{Q}_1$ Q_2 $\overline{Q}_2$	-20	-57	mA
	I_{CC} I_{CC}	3005 3005	53 54	GND GND	GND GND	GND		GND	GND GND	GND GND	GND GND	GND					GND	V_{CC} V_{CC}		30 30	
2	Same tests, terminal conditions and limits as subgroup 1, except T_A = 125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as subgroup 1, except T_A = -55°C and V_{IC} tests are omitted.																				
7 2/ 4/ T_A = 25°C			55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81	B B B B A A A A A A A B B A A A A A A B A A A A A A A	B B A B A B A A A A A A B B B A B B A A A A A B A B A	B B A A A B B B A A A A A A B A A B B A A A A A A A A	4.5 v	B B A A A B B A A A A A A A A B B A A A B A A A A A A	B B B B B B A A A A A A B B B A A A A A A A A A B B	B B B B A A A A A A A B B A A A A A A A A A A A A A	B A A B B B B A B B A A A A B B A A A A A A A A A A A	H 3/ L L H H H H L L H H H H L L H H H L L H H H H H H	H 3/ H H L L H H H H L L L L H H H H H H H L L L L L L	GND	H 3/ H H L L H H H H L L L L H H L L H H H L L H H L L	H 3/ L H L H L H L H L H L H L H L H L H L H L H L H L H	B A A B B B B A A B B A A A A B B B A A A A B A A B	All outputs	H or L as shown 3/		
8 2/ 4/	Same tests, terminal conditions and limits as for subgroup 7, except T_A = 125 and -55°C.																				

See notes at end of device type 07.

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TABLE III. Group A inspection for device type 07 [1/] - Continued.

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D / Case C / Test No.	1 / 3 / Clock 1	2 / 2 / D1	3 / 1 / Clear 1	4 / 14 / VCC	5 / 13 / Clear 2	6 / 12 / D2	7 / 11 / Clock 2	8 / 10 / Preset 2	9 / 9 / Q2	10 / 8 / $\bar{Q}2$	11 / 7 / GND	12 / 6 / $\bar{Q}1$	13 / 5 / Q1	14 / 4 / Preset 1	Meas. terminal	Test limits: Min	Max	Unit
9 T_A = 25°C	f_{MAX}	(Fig. 12)	82	IN	E	5.0 v	5.0 v							GND		OUT	5.0 v	Q1	5		MHz
			83	IN	E	5.0 v									OUT		5.0 v	$\bar{Q}1$			
			84					5.0 v	E	IN	5.0 v	OUT						Q2			
			85					5.0 v	E	IN	5.0 v		OUT					$\bar{Q}2$			
	t_{PLH}	3003 (Fig. 11)	86			IN									OUT		IN	Clear 1 to $\bar{Q}1$	5	25	ns
			87			IN										OUT	IN	Preset 1 to Q1			
			88					IN			IN		OUT					Clear 2 to $\bar{Q}2$			
			89					IN			IN	OUT						Preset 2 to Q2			
	t_{PHL}		90			IN										OUT	IN	Clear 1 to Q1		33	
			91			IN									OUT		IN	Preset 1 to $\bar{Q}1$			
			92					IN			IN		OUT					Clear 2 to $\bar{Q}2$			
			93					IN			IN	OUT						Preset 2 to Q2			
	t_{PLH}	3003 5/ (Fig. 12)	94	IN	IN	5.0 v										OUT	5.0 v	Clock 1 to Q1	5	25	ns
		(Fig. 13)	95	IN	IN	5.0 v									OUT		5.0 v	Clock 1 to $\bar{Q}1$			
		(Fig. 12)	96					5.0 v	IN	IN	5.0 v	OUT						Clock 2 to Q2			
		(Fig. 13)	97					5.0 v	IN	IN	5.0 v		OUT					Clock 2 to $\bar{Q}2$			
	t_{PHL}	(Fig. 13)	98	IN	IN	5.0 v										OUT	5.0 v	Clock 1 to Q1		33	
		(Fig. 12)	99	IN	IN	5.0 v									OUT		5.0 v	Clock 1 to $\bar{Q}1$			
		(Fig. 13)	100					5.0 v	IN	IN	5.0 v	OUT						Clock 2 to Q2			
		(Fig. 12)	101					5.0 v	IN	IN	5.0 v		OUT					Clock 2 to $\bar{Q}2$			

See notes at end of device type 07.

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TABLE III. Group A inspection for device type 07 1/ - Continued.

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D / Case C	1 / 3	2 / 2	3 / 1	4 / 14	5 / 13	6 / 12	7 / 11	8 / 10	9 / 9	10 / 8	11 / 7	12 / 6	13 / 5	14 / 4	Meas. terminal	Test limits Min	Max	Unit
			Test No.	Clock 1	D1	Clear 1	VCC	Clear 2	D2	Clock 2	Preset 2	Q2	$\overline{Q2}$	GND	$\overline{Q1}$	Q1	Preset 1				
10 T_A=125°C	F_{MAX}	(Fig. 12)	102	IN	E	5.0 v	5.0 v							GND		OUT	5.0 v	Q1	5		MHz
			103	IN	E	5.0 v									OUT		5.0 v	$\overline{Q1}$			
			104					5.0 v	E	IN	5.0 v	OUT						Q2			
			105					5.0 v	E	IN	5.0 v		OUT					$\overline{Q2}$			
	t_{PLH}	3003 (Fig. 11)	106	IN	IN	5.0 v										OUT	5.0 v	Clear 1 to $\overline{Q1}$	5	31	ns
			107	IN	IN	5.0 v									OUT		5.0 v	Preset 1 to Q1			
			108					5.0 v	IN	IN	5.0 v	OUT						Clear 2 to $\overline{Q2}$			
			109					5.0 v	IN	IN	5.0 v		OUT					Preset 2 to Q2			
	t_{PHL}		110	IN	IN	5.0 v										OUT	5.0 v	Clear 1 to Q1		39	
			111	IN	IN	5.0 v									OUT		5.0 v	Preset 1 to $\overline{Q1}$			
			112					5.0 v	IN	IN	5.0 v	OUT						Clear 2 to Q2			
			113					5.0 v	IN	IN	5.0 v		OUT					Preset 2 to $\overline{Q2}$			
	t_{PLH}	3003 5/ (Fig. 12)	114	IN	IN	5.0 v										OUT	5.0 v	Clock 1 to Q1	5	31	ns
		(Fig. 13)	115	IN	IN	5.0 v									OUT		5.0 v	Clock 1 to $\overline{Q1}$			
		(Fig. 12)	116					5.0 v	IN	IN	5.0 v	OUT						Clock 2 to Q2			
		(Fig. 13)	117					5.0 v	IN	IN	5.0 v		OUT					Clock 2 to $\overline{Q2}$			
	t_{PHL}	(Fig. 13)	118	IN	IN	5.0 v										OUT	5.0 v	Clock 1 to Q1		39	
		(Fig. 12)	119	IN	IN	5.0 v									OUT		5.0 v	Clock 1 to $\overline{Q1}$			
		(Fig. 13)	120					5.0 v	IN	IN	5.0 v	OUT						Clock 2 to Q2			
		(Fig. 12)	121					5.0 v	IN	IN	5.0 v		OUT					Clock 2 to $\overline{Q2}$			
11	Same tests, terminal conditions and limits as subgroup 10, except T_A = -55°C.																				

NOTES:
A = Normal clock pulse.
B = Momentary GND, then 4.5 v.
C = Momentary GND, then open.
D = Momentary 4.5 v, then GND.
E = Input D connected to $\overline{Q}$.

1/ Terminal conditions (pins not designated may be H $\geq$ 2.0 v, or L $\leq$ 0.8 v, or open).
2/ Input voltages shown are: A = 2.0 volts minimum and B = 0.8 volts maximum.
3/ Output voltages shall be either: (a) H = 2.4 v, minimum and L = 0.4 v, maximum when using a high speed checker double comparator; or (b) H $\geq$ 1.5 v and L $\leq$ 1.5 v when using a high speed checker single comparator.
4/ Tests shall be performed in sequence.
5/ Tests shall be performed for both D input pulses (A and B).

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(c) Lead bend in only one direction is required for initial conditioning prior to moisture resistance and salt atmosphere tests.
(d) High temperature storage test (Method 1008 of MIL-STD-883) conditions:
(1) Temperature: 150 ±10° C.
(2) Duration: 1,000 hours, except as otherwise permitted by Appendix B to MIL-M-38510.
(e) Operating life test (Method 1005 of MIL-STD-883) conditions, or equivalent:
(1) Test condition D or E, using the circuit shown on figure 4, or equivalent.
(2) T_A = 125° C, minimum.
(3) Test duration: 1.000 hours, except as permitted by Appendix B of MIL-M-38510.
(f) Omit steady state reverse bias test.

4.5 Methods of examination and test. Methods of examination and test shall be as specified in the appropriate tables and as follows.

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional current and positive when flowing into the referenced terminal.

4.5.2 Life test cooldown procedure. When devices are measured at 25° C following application of the operating life or burn-in test condition, they shall be cooled to room temperature prior to removal of the bias. Alternately, the bias may be removed during cooling if the case temperature is reduced to room temperature within 30 minutes after removal of the test condition.

4.6 Inspection of preparation for delivery. Inspection of preparation for delivery shall be in accordance with MIL-M-38510, except that the rough handling test shall not apply.

5. PREPARATION FOR DELIVERY

5.1 Preservation-packaging and packing. Microcircuits shall be prepared for delivery in accordance with MIL-M-38510.

6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for use for Government microcircuit applications (original equipment) and logistic support.

6.3 Ordering data. The contract or order should specify the following:

(a) Complete part number (see 1.2).
(b) Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.
(c) Requirement for certificate of compliance, if applicable.
(d) Requirements for notification of change of product or process to procuring activity in addition to notification to qualifying activity, if applicable.
(e) Requirements for packaging and packing.
(f) Requirements for failure analysis (including required test condition of Method 5003), corrective action, and reporting of results, if applicable.
(g) Requirements for product assurance options.
(h) Requirements for carriers, special lead lengths or lead forming, if applicable. These requirements shall not affect the part number.

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6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-STD-1313, MIL-STD-1331, and as follows:

GND - - - - Electrical ground (common terminal)

V_{IN} - - - - Voltage level at an input terminal

6.5 Logistic support. Lead materials and finishes (see .3) are interchangeable. Unless otherwise specified, microcircuits procured for Government logistic support will be procured to device class B (see 1.2.2) and lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.6 Substitutability. Microcircuits covered by this specification are substitutable for the following commercial device types:

Device type	Commercial type
01	SN5472 (Circuit A)
01	DM5472 (Circuit B)
01	MC5472 (Circuit C)
02	SN5473 (Circuit A)
02	DM5473 (Circuit B)
02	S5473 (Circuit C)
03	SN54107 (Circuit A)
03	DM54107 (Circuit B)
03	S54107 (Circuit C)
04	SN5476 (Circuit A)
04	DM5476 (Circuit B)
04	S5476 (Circuit C)
05	5474 (Circuit A)
05	DM5474 (Circuit B)
06	5470
07	SN5479 (Circuit A)
07	MC5479 (Circuit B)

Custodians:
Army - EL
Navy - EC
Air Force - 17

Review activities:
Army - EL, MI, MU
Air Force - 11, 17, 80
DSA - ES
NASA - NA

User activities:
Army - SM
Navy - CG, MC, AS, OS, SH
Air Force - 19

Preparing activity:
Air Force - 17

Agent:
DSA - ES

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

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c. Reason/Rationale for Recommendation:

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7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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